

Alcohol, Labor, and Agriculture

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I evaluate a randomized control trial in rural Kenya in which selected households received cognitive behavioral therapy and medication to reduce alcohol abuse. Sixteen months post-intervention, the program decreased the likelihood of positive spot breathalyzer tests among men by 14 percentage points, the likelihood of heavy drinking by 19 points, and increased household real annual harvest values by 32 percent. The intervention's success is likely due to a confluence of mechanisms, including expenditure reallocation, improved self-control, and greater information sharing among peers. This study contributes to growing evidence illustrating the efficacy of psychological interventions in improving economic well-being. (JEL D91, I12, I15, J22, J43, O12, Q12)

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I Introduction

Persistent high rates of alcohol abuse pose a major public health and socioeconomic burden globally. In 2019, 2.6 million deaths worldwide were attributed to alcohol consumption (WHO, 2024). Alcohol use increases the likelihood of numerous health conditions such as injuries, poisonings, hypertension, heart and liver disease, and cancer (CDC, 2021). Economically, alcohol abuse generates work-related problems including absenteeism, interpersonal conflicts, and reduced earnings, in addition to other hardships like debt.¹

Alcohol-related harm is higher in low-income countries and among marginalized populations, which further exacerbates social inequalities both between and within countries (Walls et al., 2020). Sub-Saharan Africa (SSA) has the highest global burden of alcohol-related diseases and the largest share of unregulated alcohol consumption (WHO, 2024). As part of this study, baseline data collected from 30 villages of western Kenya show that 53 percent of men consumed alcohol in the past week, with the average male drinker consuming 2.9 alcoholic beverages the last time he drank – usually large servings of informally produced liquor known as *chang’aa*. Given that more than half of SSA households are primarily engaged in agriculture (Davis, Giuseppe and Zezza, 2017), it is plausible that widespread and heavy alcohol consumption in this region may contribute not just to poor health outcomes, but also to the chronic low productivity of agriculture found across SSA countries (Wollburg et al., 2024). In response, some national governments in the region have implemented policies to reduce the heavy consumption of informal alcohol. For example, Kenya’s 2010 Alcoholic Drinks Control Act aimed to regulate and tax homemade alcohol. Unfortunately, it has faced limited success due to high costs and complex compliance procedures (Carey et al., 2015). Enforcement of regulations is weak, and homemade alcohol – sometimes adulterated with contaminants – continues to be widely available in rural regions and contributes to significant risks.

This paper examines the efficacy of an alternative policy tool to address alcohol abuse in rural Kenya using a randomized control trial: cognitive behavioral therapy

¹For example, see French et al. (2011), Jarl and Gerdtham (2012), and Böckerman, Hyytinen and Maczulskij (2017).

involving a counseling and pharmacological intervention.² I find that this intervention significantly reduced alcohol consumption, increased agricultural output, and yielded large net economic benefits. Using the empirical results and estimates from the medical literature on health impacts of decreases in alcohol consumption, I estimate total benefits of the program from extended years of life across men in the treatment to be between 523,980 and 1,047,960 USD (PPP, real 2021 terms). That conservative estimate does not include potential positive spillover effects or non-health economic benefits. By contrast, the cost of implementing the intervention was 37,327 USD.

This RCT took place in 2021 and consisted of 1050 cohabiting spouses in 525 smallholder farming households (i.e., farming around a hectare or less of land) located in Busia, Bungoma, and Kakamega counties of Kenya³ – densely populated and less developed rural areas in which agriculture is the primary economic activity. Due to inheritance practices that subdivide landholdings over generations, farm sizes are very small, often less than two acres, and farming is continuous and intensive. Poverty in these rural regions coupled with a general lack of non-agricultural work has led many men in this areas to migrate to larger towns for employment, leaving the rest of the household behind in the village. For those remaining in these villages, entrenched poverty is associated with both the supply and demand of alcohol: cottage industries producing alcohol can provide a supplementary source of income, while the poor spend a relatively high share of their income on alcohol and tobacco (ranging from 1 to 8 percent of their budget, depending on the country (Banerjee and Duflo, 2007)). This can, in turn, lead to higher poverty (Jayathilaka, Selvanathan and Bandaralage, 2016).

Individuals in households randomly placed into the treatment group were invited to participate in facilitator-led counseling sessions modeled after “Motivational Enhancement Therapy” (MET) to aid in the reduction of alcohol consumption (Miller and Rollnick, 1992). This program was composed of several parts: first, after initial contact and introduction to the implementing organization, the individuals

²The project was preregistered in the AEA RCT Registry #AEARCTR-0006413 (Murphy and Castilla, 2022). This paper analyzes a subset of the project’s outcomes.

³There are 47 counties in Kenya and they are the highest regional administrative district in the country since 2010. They are similar to U.S. states, though typically smaller in size.

in treatment households were invited to participate in three full days of workshops and discussions with village peers, utilizing peer effects to incentivize sobriety. Second, they were invited to take part in numerous personal and spousal counseling sessions – each for several hours over two weeks – designed to motivate commitment to decrease drinking and engender trust, communication, and accountability between spouses. Third, as a pharmacological component, a nurse practitioner prescribed vitamins to all treated individuals and medication to those experiencing withdrawal symptoms. Enumerators then conducted surveys with sampled individuals an average of 6 weeks and, in the second round, 16 months after the conclusion of project activities in a particular village. Selected households must have had at least one spouse who was an alcohol consumer in the baseline survey, but otherwise are not screened based on quantity of alcohol consumed nor on past or current motivation to decrease alcohol consumption.

I use three methods to measure alcohol consumption: through a self-reported alcohol consumption survey to individual i , by privately and concurrently asking individual i 's spouse about i 's alcohol consumption, and through unannounced spot breathalyzer tests. Additionally, as the sample is composed entirely of agricultural households, I collect data on agricultural inputs and yields, as well as consumption expenditures and labor market outcomes. I compare post-program and baseline survey data from treated households to two separate control groups. The “pure control” consists of households in non-treatment villages. The “spillover control” consists of households in treatment-villages who were not randomly selected to participate in the program. Given close and interconnected relationships among households within villages in this region, it is likely that effects arising from any intervention would spread from treated households to these spillover controls. For example, if a critical mass of regular drinkers assigned to the treatment reduce alcohol consumption, this endogenous information signal could encourage self control among others not assigned to the treatment (Battaglini, Bénabou and Tirole, 2005).

I find that the program had large and statistically significant effects on alcohol consumption among men and agricultural outcomes. The medium-run intent-to-treat (ITT) results (collected in 2022, approximately 16 months after the intervention) indicate that the treatment decreased the likelihood of spouse-reported alcohol

consumption among men in the week prior to the survey by 19 percentage points (pp) ($p < 0.01$), an approximate 22 percent decrease from the pure control. I also find a 19 pp decrease in the likelihood of the husband being a heavy drinker (defined for men as consuming 15 or more drinks in a week) according to their spouse ($p < 0.01$; a 29 percent decrease relative to the pure control). Furthermore, an enumerator conducted unannounced spot checks of men in the original sample using a breathalyzer in 2023 – more than two years post intervention. Analysis of these data show that men in the treatment are 14.3 pp less likely to test positive for alcohol consumption compared with those in the pure control group ($p < 0.001$), representing a 40 percent lower likelihood and providing objective evidence of the longer-run impact of the program on alcohol consumption.

In addition, by the time of the medium-run survey, I find evidence of within-village spillover effects of the intervention on alcohol consumption. According to their wives, men in the spillover control are 12 pp less likely to have drunk in the past week ($p < 0.05$) and 13 pp less likely to be a heavy drinker ($p < 0.05$) relative to men in the pure control. Moreover, while of a lower magnitude, these results are statistically indistinguishable from the medium-run effects on men in the treatment. This is suggestive evidence that peer effects and shifting social norms within the village may have amplified the intervention's impact over time to those beyond the original treatment sample.

Household-level analysis suggests that the intervention also had a significant effect on agricultural outcomes. Controlling for pre-intervention agricultural data and using GPS measured plot sizes, results indicate that assignment to the treatment increases total annual (i.e., two season) inorganic fertilizer use by 31 percent in the year following the intervention ($p < 0.01$). Real annual harvest values increase by 32 percent ($p < 0.01$), or about 11,570 Kenyan Shillings (KSh) (that is, 102 USD at the time of the intervention). Suggestive evidence points to a shift among individuals assigned to the treatment from wage labor to self-employment. However, insufficient power precludes identification of statistically significant effects related to labor market outcomes.

A systematic evaluation of causal pathways linking the intervention with these positive economic outcomes suggests numerous potential mechanisms. Reductions

in alcohol use, for example, may have led to a reallocation of spending from alcohol to agricultural inputs. Although most results from an expenditure analysis are not robust to multiple hypothesis testing, I find that after 16 months, spending on alcohol in the month prior to the survey is 48 percent (631 KSh) less among men in the treatment group compared to the pure control. Moreover, as compared to the pure control group mean, assignment to the treatment increases monthly agricultural expenditures on inputs and livestock by 82 percent. Other potential mechanisms derive from the group workshops, which were in part designed to create platforms of support among households assigned to the treatment. I find those assigned to the treatment are more likely to get help decreasing alcohol consumption and more likely to receive agricultural advice from others assigned to the treatment relative to peers in the control group. Still other potential mechanisms stem from greater peace and cooperation in the household due to reductions in alcohol use leading to drops in intimate partner violence.

From an economic perspective, heavy alcohol use in general is concerning as it can lead to negative effects on earnings, employment, and other related outcomes. For example, French et al. (2011) show that alcohol abuse increases the likelihood of employment problems through greater interpersonal conflicts in the workplace, while Jarl and Gerdtham (2012) find heavy alcohol consumption increases the risk of long-term sickness-related work absences. Böckerman, Hyytinen and Maczulskij (2017) show former and heavy drinkers have about 20 percent lower earnings than moderate drinkers. Greater access to alcohol can also lead to impulsive consumption and higher debt through a relaxing of implicit commitment devices (Ben-David and Bos, 2021). In a developing country context, studies suggest causal links between alcohol consumption and intimate partner violence (Luca, Owens and Sharma, 2015; Dixit and Mukherjee, 2025), which could have spillovers to household economic outcomes. Moreover, spending on alcohol and tobacco can crowd out spending on consumer durables and foodgrains (Jumrani and BIRTHAL, 2017), a particular problem in SSA given high levels of rural poverty plus alcohol consumption that is 24 percent greater than the global average (WHO, 2024).

Despite these clear costs, relatively few economics studies evaluate the efficacy of interventions designed to decrease alcohol consumption, particularly for

developing countries. Within India, Khurana and Mahajan (2022), Chaudhuri et al. (2024), and Luca, Owens and Sharma (2015) find that restrictions on the sale of hard liquor and other alcohol prohibitions reduce violent crime, sexual assault, and aggregate violence against women. Law and Marks (2020), Barron et al. (2024), and Nakaguma and Restrepo (2018) in the U.S., South Africa, and Brazil, respectively, find alcohol prohibitions significantly decrease mortality. In one of the few randomized control trials related to alcohol abuse in a less developed country, Schilbach (2019) finds strong demand among Indian rickshaw drivers for commitment devices to reduce alcohol consumption but small and statistically insignificant effects on earnings and hours worked. Surprisingly, he finds negative effects on labor supply at the extensive margin. Overall, this points to a need to determine whether a different form of substance abuse intervention coupled with longer follow-up periods would identify positive economic impacts.

This study also adds to the growing literature analyzing the effects of counseling and therapy on economic outcomes. Bossuoy et al. (2022) in Niger demonstrate that interventions that included psychosocial components were most effective in helping extremely poor women build sustainable sources of income. Baranov et al. (2020) find that seven years after a psychotherapy intervention among clinically depressed mothers in rural Pakistan, women in the treatment had better mental health, more financial empowerment, and greater time and money-intensive parental investments relative to women in the control. In a meta-analysis, Lund et al. (2024) find that psychosocial interventions conducted across 39 interventions lead to significant average improvements in various outcomes, such as a reduction in the work absences and increases in wealth.

Perhaps closest to this study, Nadkarni et al. (2017) report results of a randomized control trial in Goa, India, comparing those receiving standard alcohol treatment in a primary healthcare setting to those who were also treated with MET similar to the protocol used in this study's intervention. After three months, the researchers found that the share of patients in remission among those receiving the counseling was 10 pp higher ($p = .01$), and those abstaining from alcohol in the past 14 days was 24 pp higher ($p < 0.001$), than the share of those receiving only the standard form of care. The authors note, however, that among those who continued

to drink, the mean level of alcohol consumed was not statistically different between treatment groups. This present study diverges from this and other previous work in this area by implementing an RCT related to substance abuse in a rural African setting and using a 16-month follow-up period. Moreover, this study’s sample includes any household with an alcohol consumer, rather than those who are already medical patients due to their excessive alcohol consumption.

The remainder of the paper is organized in the following way. In Section II, I describe the background of alcohol consumption in SSA and describe the setup of the randomized control trial, sample selection, balance, and attrition. I then present the empirical methods and detail the results on alcohol consumption using spouse reports and breathalyzer data, agricultural outcomes, labor, and wages in Section III. After presenting these results, I discuss potential mechanisms in Section IV. In Section V, I discuss the net benefits of the program, and I conclude in Section VI with a discussion of policy implications and future avenues of research.

II Experimental Design

The sample comes from 30 randomly selected villages of western Kenya (Murphy, 2025a).⁴ The villages are very characteristic of the rural areas of this region of Kenya – densely populated, poor, and consisting primarily of smallholder farmers. Because a portion of the intervention (discussed in detail in Appendix B) relies on spousal counseling and a major interest is in intrahousehold economic outcomes, the sample is limited to cohabiting spouses. Moreover, due to safety protocols (also discussed in Appendix B), this project only includes households in which both spouses were 55 years old or younger and without a serious, long-term illness. In each village, project staff contacted the chief or a village elder and they prepared village lists of all the households that satisfied these conditions. For the baseline, the enumerators attempted to survey both spouses in all these households, resulting

⁴In some of these villages, the International Institute of Tropical Agriculture has previously worked on unrelated agricultural projects that were in locations that were effectively random, conditional on being rural areas of western Kenya. The remainder of the villages were randomly chosen for this study from rural areas in Bungoma, Kakamega, and Busia counties.

in 1,835 surveys.⁵ There were no instances of individuals refusing to participate in the baseline survey. I then exclude individuals who have serious chronic illness (16 individuals) older than 55 (36 individuals), or who did not have a spouse currently cohabiting with them or where the spouse was in the two previous categories (225 individuals). This leaves a sample of 1,558 individuals (779 households).

The baseline survey, which began in September 2020, was short and consisted of basic demographic questions, questions related to the amount of time worked in various capacities, alcohol consumption, and questions related to intimate partner violence.⁶ Of the 1,558 individuals, 45.8 percent report having consumed alcohol in the past 30 days (64.8 percent of men). Households without drinkers in the baseline survey were then excluded from further participation in the program (508 individuals).⁷ The final sample is 525 households (1050 individuals). It is worth noting that these households with drinkers were not selected based on interest or motivation to reduce drinking but simply contain an alcohol consumer.

The randomization proceeded in two stages. First, 10 of the 30 villages were randomly allocated to the pure control group, and villages in this category received no intervention, whereas the remaining twenty villages are referred to as treatment villages. I include a map of treatment and control village locations in Appendix Figure A1. As the figure shows, the villages are distributed over a wide area of western Kenya, and there is adequate spatial distribution of both the treatment and control villages. I then randomly assign households in the twenty treatment villages into either a treatment group or a spillover control group. This household-level randomization was stratified at the village level, such that approximately 55 percent of each treatment village was allocated to the treatment group, and 45 percent to the

⁵We do not have data on the share of households in each village that this represents. Polygamous households exist in this sample (approximately eight percent of total). For these households, enumerators interviewed the man and senior wife. In households where there were actually only one adult present, that adult was interviewed in this baseline.

⁶Results of the program with respect to intimate partner violence are discussed in a companion paper, Castilla, Aqeel and Murphy (2025).

⁷There is variability in the share of households in a particular village that have an alcohol consumer, ranging from 44 percent to 92 percent (median and mean across villages of 64 and 67 percent, respectively). Villages in Kakamega county have the highest share of households with an alcohol consumer as a percent of the total (where the difference is statistically significant at the five percent level).

spillover control. The randomization leads to 328 individuals in the pure control, 322 in the spillover control, and 400 individuals in the treatment group. Appendix Figure A2 provides an illustration of the project’s randomization.

In general, the project’s randomization appears to have been effective. Appendix Table A1 shows direct balance checks on variables from the baseline survey. I include three specifications for each variable: the first with the baseline sample (1050 individuals), the second with the sample from a 2021 post-intervention short-run survey (982), and the third column with the sample from the medium-run survey (899). These tests show there is no imbalance on baseline variables across survey rounds.

Project staff implemented the program in each treatment village in several stages, incorporating a bundle of services including both workshops and discussions together with subsequent counseling sessions conducted individually and jointly with both spouses. The goal of the implementing organization was to achieve non-use of alcohol within the household, and the intervention itself was designed using a framework after Miller and Rollnick (1992) known as “Motivational Enhancement Therapy” (MET). This counseling technique involves assessing the person’s drinking habits and providing feedback about their effect on health and social/family relationships. MET focuses on creating a strong therapeutic relationship between counselor and addict and highlighting any discrepancies between their current behavior and desired goals (SAMHSA, 2019). The counselor encourages the individual to enumerate their own motivations and commitment to change, collaborating on a plan to decrease alcohol consumption and prevent relapse. Within its application in this project, counselors also utilized the “Stages of Change” framework (Prochaska and Di Clemente, 1982), and using the Stages of Change Readiness and Treatment Eagerness Scale (SOCRATES) (Miller and Tonigan, 1996), established the respondent’s baseline readiness for change, and over several weeks worked towards increasing this level. Studies show that these tools are effective aids in treating alcohol addiction (see review in Rosenthal et al. (2007)).

Beginning in February 2021, the implementing organization began implementing the alcohol remission intervention in the twenty treatment villages.⁸ Those ran-

⁸The implementing organization for the project intervention, the Giselle Foundation, is an offi-

domly assigned to the treatment were invited to participate first in group workshops, followed by individual and spousal counseling sessions. The group workshops took place over three full days in a central location in the village and consisted of an education component – counselors would teach attendees about the negative effects of alcohol on health, relationships, finances, etc. – followed by discussion activities. Groups sessions were designed to create platforms of mutual support among those facing the negative effects of alcohol in each treatment village. Following group workshops, counselors would visit the homesteads of treatment households to conduct individual and spousal counseling.⁹ The goal of the intervention team was to complete eight sessions with each household in which both spouses were present. The teams sought to work around the schedules of the households, but in some cases were not able to complete all eight due to the inability (or lack of interest) of a spouse to find time to attend the session. Appendix Table A2 shows the number of sessions completed in the sample. A total of 79 of 400 individuals assigned to the treatment did not participate in counseling sessions. The mode number of sessions that a treatment individual participated in was eight, which represents 41.5 percent of the treatment sample.¹⁰ This stage was the longest phase of the intervention, and lasted approximately two weeks in each village. I include additional details of the intervention, along with a description of modest participation incentives (expected value of approximately a day’s wage) and safety protocols in Appendix B.

A Timeline and Data

The baseline survey for this project began in September, 2020 and lasted through December, 2020, in which enumerators surveyed villages and households within villages in a random order. In 2021, the local partner organization implemented the intervention in treatment villages in a random order between February and May and enumerators meanwhile conducted the short-run surveys from February through

cially non-religious Kenyan NGO based in Kisumu, Kenya.

⁹Appendix C contains anonymized excerpts from counselor notes relating to this strategy.

¹⁰Appendix Table A2 also shows that there are individuals who received more than eight counseling sessions. In these cases, the intervention teams were seeking to have joint sessions with both spouses, but had found only one spouse at home. That one spouse would thus receive a session, but the counselors would return for make-up sessions when both spouses were available with the goal of eight sessions completed with both spouses.

early July in control villages and in treatment villages as the intervention concluded. In post-intervention surveys, household-level data were collected from husband and wife jointly, with individual-level data collected privately and concurrently from each spouse by two enumerators who matched the gender of the respondent.

While the intervention and follow-up surveys happened in the context of COVID-19, rural western Kenya was relatively unaffected by policy responses. In Kenya, lockdowns only affected major cities and other restrictions such as curfews and limits on social gatherings had expired prior to the start of the project's interventions. These lockdowns, which temporarily prevented economic migration from villages, together with limits on cross-border trade with Uganda affected the economy in this region to some extent. However, because this is a randomized trial, I assume these economic effects of COVID-19 were similar between treatment and control villages. To the knowledge of the project implementers, neither illness nor policy responses related to COVID-19 affected the intervention, participation, or data collection in any way and should not impact external validity.

In order to gauge the medium-run impacts of the program, enumerators attempted to survey the same households between May and October 2022. Enumerators surveyed all villages (including pure control villages) in a random order, with the average household in a treatment village receiving the medium-run survey 16 months after the completion of the intervention in that village. In Appendix Figure A3, I include a histogram of the duration between the end of the intervention and this medium-run survey, which shows an approximately normal distribution of months between when the intervention concluded in a treatment village and when the individual received the medium-run survey. In Appendix Table A3, I show the distribution of surveyed individuals by treatment assignment and survey month in the medium-run survey. In the remainder of this paper, I refer to this survey as the 16-month post intervention, medium run, or 2022 survey, for convenience.

To obtain an objective source of alcohol consumption data and to analyze the longer-run effects of the program, an enumerator conducted unannounced spot breathalyzer checks of men in the sample from November, 2023 to January, 2024 (participants were not tested immediately before, during, or after Christmas and

New Year holidays).¹¹ An enumerator visited villages in a random order as predetermined by a random number generator. A total of 414 men had their blood alcohol content (BAC) measured using a breathalyzer, while an additional nine men from the sample refused the breathalyzer.¹²

Western Kenya has two agricultural cropping seasons per year: the “long-rains,” in which planting is usually in March-April and harvesting is in August-September, and the “short-rains,” in which planting is in September-October and harvesting in January-February. During the 2021 and 2022 surveys, enumerators asked the household head questions related to their agricultural production on their primary farming plot over the past two fully complete (i.e., harvested) agricultural seasons. This means that each of these surveys contains agricultural data of one season of short-rains and one season of long-rains production for each household. In the 2021 short-run survey, all completed agricultural seasons were from *prior* to the intervention, meaning these data provide an agricultural baseline.¹³ Due to the timing of the medium-run survey, some farmers in the 2022 survey round were asked about different seasons depending on whether they had completed harvesting their most recent crops. In estimations with agricultural outcomes I incorporate a control to identify whether a farmer had fully harvested their 2022 long-rains crops and thus indicating which seasons for that household are included in the data.

I include summary statistics for alcohol consumption and other outcome variables in Table 1. Men in the sample on-average drink considerable levels of alcohol: in the medium-run survey, spouse-reported data show 75 percent of men in the sample drank in the past week, with an average consumption of 22.3 US standard drinks (about 3.9 servings of chang’aa).¹⁴ Moreover, 55 percent of men can

¹¹While unannounced, consent was obtained immediately prior to giving the breathalyzer. This is an added component of the project and is not contained in the RCT preregistration.

¹²The specific breathalyzer used was a BACtrack trace professional breathalyzer, with the manufacturer stating that the sensor accuracy is +/- 0.005 at 0.050 percent BAC.

¹³Timing of the 2021 short-run survey does not influence these data, as all households were asked about the same, pre-intervention agricultural seasons.

¹⁴I use the following conversion factors to compute liters of ethanol and US standard drink equivalents: chang’aa is 34 percent ethanol, busaa is 4 percent ethanol, beer is 5 percent ethanol, wine is 12 percent ethanol, and commercial spirits are 40 percent ethanol. Percent ethanol for locally produced drinks confirmed as reasonable approximations by field tests. These field tests find that the average serving size of a glass of chang’aa is 300 milliliters, and serving sizes of busaa are

be considered heavy drinkers in the medium run (at least 15 US standard drinks consumed in the past week), and the average maximum number of standard drinks consumed in one day in the past week is 5.1. Notably, the spouse-reported figures indicate significantly more alcohol consumed by her husband than using self-reported figures, which may indicate bias in self-reported data. For this reason, the alcohol-related findings report results using spouse-reported data and from spot breathalyzer checks.

With regard to alcohol consumption and hours worked, there were some changes in questions between survey rounds. In the baseline survey, enumerators asked whether the respondent drank 1) in the past month, 2) in the past week, and if so, 3) how many alcoholic drinks were consumed the last time they drank in the past week. The 2021 and 2022 surveys investigated alcohol consumption in significantly more detail, collecting data on type and quantity of alcohol consumed by both themselves and their spouse over the past week, as well as asking for the most alcohol they consumed in a single day during the past week as an indicator of binge drinking. With respect to hours worked, the baseline survey asked about hours worked in the past week in four categories: own-farm, a different farm, non-farm wage work, and housework. The 2022 survey increased the detail of these questions and added a time-use module after Field et al. (2023), in which respondents were asked to divide up their last 24 hours into various categories.

Attrition was relatively low and balanced across survey rounds. Appendix Tables A4 and A5 show tests for differential attrition at the individual and household-level, respectively. Attrition was 6 percent between the baseline and short-run survey, 14 percent between the baseline and medium-run, and 19 percent between the baseline and breathalyzer tests. In each round, attrition was balanced across treatment status and most other baseline indicators. An exception is age, which was imbalanced ($p < 0.01$) between the baseline and medium-run survey rounds. The

typically 0.5 or 1 liters. A US standard drink is assumed to contain 17.75 ml (0.01775 liters) of ethanol. Enumeration staff collected data on serving size along with quantity of drinks consumed in a given time period. The intensive margin outcome variable for alcohol consumption thus uses these detailed data that have been converted to US standard drink equivalent consumed in a given time period, whereas the baseline variable is the number of alcoholic drinks consumed the last time they drank in the past week, as reported by the respondent.

most likely reason for this imbalance is the migration of young men from villages to larger cities for work, a common occurrence in these areas, and economic migrants tend to be younger. Other reasons for attrition include the death of an individual, or spouses who had separated/divorced. In a couple cases, the entire household had moved locations. In these instances, enumerators surveyed them at their new location and kept them in the sample. Because attrition is not perfectly balanced across all variables, I calculate bounds on estimated intent-to-treat effects after Lee (2009), which are presented on estimation tables as described below.

III Empirical Methods and Results

The primary estimation strategy uses a fully interacted model, or analysis of heterogeneous covariance (ANHECOVA), which Ye et al. (2023) demonstrate achieves greatest efficiency across a wide spectrum of randomization procedures. There was imperfect compliance among those assigned to the treatment, and I thus estimate intent-to-treat (ITT) effects to examine the effect on assignment to the treatment. This should lead to estimates that are more policy-relevant, as any program of this nature will have imperfect compliance among those assigned to the treatment. The primary estimation is therefore given by the following:

$$Y_{it} = \alpha + \beta_1 T_h + \beta_2 S_h + \sum_n X'_{in} \gamma_n + T_h \sum_n (X_{in} - \bar{X}_n)' \delta_n + S_h \sum_n (X_{in} - \bar{X}_n)' \zeta_n + \varepsilon_i \quad (1)$$

where Y_{it} is an individual i 's outcome in the medium-run and T_h and S_h are indicators of treatment and spillover control assignment of the household h , respectively, which was randomly assigned after the baseline and prior to the intervention (households assigned to the pure control are the omitted group). The vector X_i consists of the baseline version of the outcome (e.g., measure of baseline alcohol consumption) as well as other necessary controls (e.g., gender in pooled estimations, plot area in agricultural estimations), and $\bar{X}$ its sample mean. The coefficient β_1 represents the ITT effects and is the primary coefficient of interest. Coefficient β_2 , meanwhile represents the spillover effects of the intervention. In all estimations I calculate heteroskedasticity-robust standard errors clustered at the village level.

A Spouse-reported alcohol consumption

I present main results of the effect of the intervention using spouse-reported alcohol consumption in the week prior to being surveyed in the top panel of Table 2, in which I show results from fully-interacted ANHECOVA estimates (I include ANOVA and ANCOVA estimates in Appendix Table A6, which show similar results). The results indicate a significant effect of the program on alcohol consumption both in the short and medium run. In the short run, assignment to the treatment decreases the likelihood of alcohol consumption by 17.7 pp compared to the pure control (Columns 1). The magnitude of the impact is larger among men (30.3 pp, Column 2, $p < 0.001$) and smaller among women (4.3 pp, Column 3) – unsurprising given that relatively few women drink in this sample.¹⁵ I find that these effects attenuate in magnitude in the medium run (i.e., 16 months post-intervention). Overall, in the medium run assignment to the treatment decreases the likelihood of drinking in the past week according to the spouse by 12.2 pp at the mean (Column 4; $p < 0.01$), 19.0 pp among men (Column 5; $p < 0.01$), and 5.3 pp among women (Column 6). Moreover, attrition bounds are tight, with estimated bounds on the effects among men in the medium run ranging from a 18.9 to a 19.7 pp decrease in the likelihood of drinking.

I also find evidence of spillovers from those assigned to the treatment into other households in the treatment villages. While in the short run I can reject the null hypothesis of equivalence of estimated effects between the those assigned to the treatment and spillover control, in the medium-run I cannot. In addition, the medium-run effect among men assigned to the spillover control group (11.7 pp lower likelihood of consuming alcohol) is large and statistically significant (Column 5; $p < 0.05$). This is suggestive evidence that over time, the effects have diffused outside those assigned to the treatment.

Of particular concern to policymakers and those engaged in public health advocacy is heavy drinking, defined as at least 15 standard drinks in the past week for men (8 for women), which can have serious and long-term negative effects on

¹⁵At baseline, 27 percent of women in the sample drank in the past week relative to 80 percent of men – a difference significant at less than the one percent level.

health (Wood et al., 2018).¹⁶ I show the results of the intervention on likelihood of heavy drinking on the bottom panel of Table 2. According to spouses, 31 percent of those sampled in the pure control group were heavy drinkers in the short run – 56 percent of men, which increased to 66 percent of men in the medium run.¹⁷ I find that overall in the short run, those assigned to the treatment are 14.5 pp less likely to be a heavy drinker relative to the pure control (Column 1; $p < 0.001$), whereas men are 27.4 pp less likely (Column 2; $p < 0.001$) and women 2.1 pp less likely (Column 3). Similar to extensive margin results, the magnitude of the estimated effects fall in the medium run, but are still large and statistically significant overall and for men. I find that across men and women, assignment to the treatment decreases the likelihood of heavy drinking by 11.2 pp (Column 4, $p < 0.01$), with the magnitude higher for men (19.3 pp decrease; Column 5; $p < 0.01$) and smaller for women (3.9 pp decrease; Column 6). Similar to above, attrition bounds are tight, with the estimated magnitude of the effect for men assigned to the treatment falling between 19.1 and 19.7 pp.

I also analyze the results of spouse-reported alcohol consumption at the intensive margin and present results in Table 3, where the top panel shows the effect on number of drinks and the bottom panel on the maximum number of drinks consumed in one day in the past week. I estimate the effects at the intensive margin using Poisson QMLE, which is applicable to non-count data (Santos Silva and Tenreiro, 2006) and remains consistent under overdispersion (Arvis and Shepherd, 2013). Moreover, Chen and Roth (2024) show Poisson QMLE is a robust alternative to “log like” transformations when outcomes contain zeroes. The estimated effects for intensive margin results can be interpreted as a percent change using the incidence rate ratio ($e^\beta - 1$). I calculate intensive margin bounds following Lee (2009) and Chen and Roth (2024), trimming the pure control group separately in each time frame at the top and the bottom of the distribution to account for the share

¹⁶In the project preregistration, alcohol consumption was a specified outcome. However, heavy drinking in itself was not a preregistered outcome. In this paper, I define it according to standards in the medical literature of at least 8 drinks in a week for women and 15 for men (Dawson, Grant and Li, 2005).

¹⁷As noted above, baseline variables and attrition were generally balanced across treatment rounds, so this difference does not arise from a changing sample composition. I also include attrition bounds of estimated intent-to-treat effects in the table for reference.

of “always-takers” in the sample, who would drink regardless of treatment. Doing this requires the assumption that anyone who drank when assigned to the treatment would also have drunk without treatment assignment. This seems a reasonable assumption, however, given the intervention was designed to decrease drinking.¹⁸ Point estimates suggest that the treatment reduced drinking by 70 percent in the short run (top panel, Column 1; $p < 0.001$) and 44 percent in the medium run (top panel, Column 4; $p < 0.001$) corresponding to 26 and 14 US standard drinks, respectively. However, intensive margin bounds are wide – ranging from a short-run decrease of 27 to 78 percent to an 24 to 54 percent decrease in the medium run.

Furthermore, I find that assignment to the treatment led men to drink 67 percent fewer drinks in one day in the short run (corresponding to 6.4 standard drink decrease relative to the pure control mean), with the medium-run magnitude falling to 36 percent (2.4 fewer standard drinks; bottom panel of Table 3, Columns 1 and 4, respectively). Intensive margin bounds are similarly wide for this outcome, ranging from a short-run reduction of 32 to 76 percent to a medium-run decrease of 7 to 47 percent.

As I will discuss in more detail below, these results are consistent with data from surveys regarding self-reported expenditures. I find that spending on alcohol in the medium run falls by 48 percent, or approximately 631 KSh in the month prior to the 2022 survey (158 KSh in a week) when compared to the pure control group ($p < 0.01$). A serving size of chang’aa costs approximately 100 KSh, and total alcohol consumption in one week declines by between 24 and 54 percent (7.5 and 16.8 US standard drinks; top panel, Columns 5 and 6, Table 3), which is equivalent to between approximately 1.3 and 2.9 servings of chang’aa (valued at 130 to 290 KSh). The self-reported fall in alcohol expenditures are thus within the bounds of the decrease in spouse-reported alcohol consumption.

B Breathalyzer results

In addition to alcohol consumption data collected from spouse and self reports, an enumerator visited villages in a random order and conducted unannounced spot

¹⁸Using this methodology, intensive margin results for women cannot be calculated given the prevalence of zeroes in the data (i.e., most women are nondrinkers).

checks for alcohol consumption among men in the sample beginning in November, 2023. This was an added component to the project and was not included in the preregistration. Nine men in total refused the breathalyzer tests, and these men are counted as having consumed alcohol for this analysis (observations from the enumerator indicate that these men were indeed intoxicated). Most men were given the breathalyzer in the afternoon or early evening, and I control for the time of the test, and whether the test was conducted on a weekend, in the estimations.

Estimates from the linear probability model in Table 4 provide objective evidence for the longer-run effects of the intervention on alcohol consumption. Analyzing the fully-interacted ANHECOVA estimates (Column 3), assignment to the treatment leads to a 14.3 pp ($p < 0.001$) lower likelihood of a positive test result for alcohol consumption during the spot check compared to those in the pure control (in which 36 percent register as having recently consumed alcohol). Other estimation methodologies (i.e., ANOVA and ANCOVA in Columns 1 and 2, respectively) find similar effect magnitudes.

These breathalyzer test results can be compared to the top panel of Table 2, in which women in households assigned to the treatment reported their spouses to be 19 pp less likely to have consumed alcohol in the past week than those in the pure control group. While spouse-reported drinking and the findings using the breathalyzer are fundamentally different measures (the former measures whether there was consumption of alcohol over the past week while the latter measures consumption in the past day), objective data from the breathalyzer generate results consistent with the spouse-reported data and suggest durability of the program effects.

A natural question arises, however, given the differences in the relative size of the effects compared to pure control means between the spouse-reported and breathalyzer test results. The former suggests medium-run effects that are 22 percent of the mean while the latter indicates a relative effect size of 40 percent. The difference is likely due to the different measurement period: the spouse-reported data report results from a seven-day recall whereas the breathalyzer only measures drinking within the past day. The likelihood of alcohol consumption in the spouse-reported result will be higher given that drinking among occasional alcohol consumers is included in that average, whereas the probability of alcohol being detected on the

breathalyzer on that particular day the spot check is conducted is more likely among frequent drinkers. It is thus possible to interpret the results to represent the effect among “any drinkers” (86 percent of men in the medium-run pure control) and “frequent drinkers” (36 percent of men in the pure control). These differences in the measurement period thus drive differences in relative effect sizes.

C Agricultural and labor outcomes

I analyze the effect of assignment to the treatment on four agricultural indicators collected in the 2022 survey from the primary farming plot: annual (i.e., two season) real harvest value, annual fertilizer application, annual organic input application, and annual number of laborers hired (the latter two outcomes were not included in the preregistration). In agriculture, labor, fertilizer, and organic matter are key inputs for production. On plots with continuous cultivation as is most common among smallholder farmers, (inorganic) fertilizer is required to replenish soil nutrients such as nitrogen and phosphorus that otherwise would be depleted through crop cultivation. Organic inputs, such as crop residues, manure, and compost, are usually also necessary for healthy and productive soils by improving soil structure, fertility, moisture retention, nutrient availability, and microbial activity while reducing erosion and maintaining healthy soil pH (Kimetu et al., 2008; Mateete, Nteranya and Woomer, 2010). While fertilizer use is common in western Kenya, farmers often do not apply optimal amounts nor do they typically apply sufficient organic inputs (Murphy et al., 2020). Indeed, in SSA, organic inputs are often applied as a substitute for fertilizer (Sheahan and Barrett, 2017), whereas applying them as complements is most optimal for high crop yields (Vanlauwe et al., 2010).

To ensure demand effects are not driving these self-reported agricultural data given that the baseline recall data was collected post intervention, I first estimate the effect of assignment to treatment on these baseline data. The results in Appendix Table A7 reveal a baseline imbalance only with respect to organic input usage, where those in the treatment report significantly less used than those in the pure control. These baseline results mitigate concerns that those assigned to the treatment are over-reporting agricultural outcomes due to their treatment assignment. I control for these baseline outcomes when estimating the main results in

ANCOVA and ANHECOVA estimations.

I show the primary results of the agricultural analysis in Table 5.¹⁹ The fully-interacted ANHECOVA estimates suggest that treatment assignment increases annual harvest values by 32 percent at the mean (Column 3; $p < 0.01$), representing an increase of about 11,570 KSh (approximately 102 USD at the time of the survey) relative to households in the pure control. Moreover, I find households assigned to the treatment increase annual fertilizer use by 31 percent (Column 6, $p < 0.01$). The results for harvest values and fertilizer use are each similar to results estimated using ANOVA and ANCOVA (Columns 1-2 and 4-5, for each outcome, respectively) and robust to multiple hypothesis corrections. With regard to organic inputs, however, I cannot identify a statistically significant effect due to the high variance of the data. While the estimated magnitude of 33 percent is large and negative (Column 9), farmers assigned to the treatment at baseline begin with much lower organic input use (see Appendix Table A7). Given most organic inputs are produced *in situ* in the form of crop residues and farmyard manure, increasing use of organic inputs can take significantly longer than other input types. In contrast, treatment assignment increases the amount of hired labor by 53 percent at the mean using the preferred ANHECOVA estimates (Column 12, $p < 0.05$), although we should take this result with some caution as the magnitude is smaller and statistically insignificant in the ANOVA and ANCOVA estimates (Columns 10 and 11).

On Table 6, I present ANHECOVA estimation results analyzing the effect of the intervention on hours worked and wage earnings at the individual level (top and bottom panels of the table, respectively). I include analogous ANOVA estimates in Appendix Table A8). Insufficient power and high variance in the outcome data prevent identifying any statistically significant results, although some estimated treatment effects are of relatively high magnitude and warrant consideration.

With regard to time worked, the outcome variables represent hours of labor in the last 24 hours on: 1) their own farm, 2) in self-employed income generating activities (e.g., working in a shop they own, driver on motorbike they own, etc.), or 3) in income generating work for wages (e.g., working on someone else's farm,

¹⁹These results and those for hours worked, wages, and expenditures below are estimated using a Poisson QMLE model with heteroskedasticity-robust standard errors clustered at the village level.

school teacher, etc.). For hours worked on own farm, the pooled gender sample and estimates among men are small and near zero, but I find relatively high, positive magnitudes for women (top panel, Column 3), suggesting a 15 percent increase in hours worked on the farm (minimum detectable effect (MDE) among women for 5 percent statistical significance is 66 percent). Increases in hours in self employment is positive and relatively large across both the gender pooled and gender sub-samples, suggesting an overall increase of 22 percent (Column 4; MDE of 58 percent). Meanwhile, I find large decreases in hours worked for actual wages (Columns 7 through 9), suggesting that the treatment may have contributed to a shift from wage earnings to self employment, although these results are not statistically significant (MDE of 37 percent in the pooled sample).

In the bottom panel of Table 6, magnitudes for the extensive margin of selecting into any wage earnings (Columns 1-3) and non-farm wage earnings (Columns 4-6) are generally small and near zero. The exception may be women selecting into any wages, with the estimated effect of the treatment decreasing the likelihood of working for wages by 11 pp (MDE of 16 pp). I also find a large decrease in total labor earnings for women (24 percent; Column 9; MDE of 37 percent). As I further discuss below, this may be due to women substituting wage labor to joining husbands staying busy on the farm to aid sobriety.

D Randomization inference

As a robustness check, I use randomization inference to test whether the results are sensitive to finite sample bias or through any potential treatment group imbalance. I conduct again the two stages of randomization (at village and household level) one thousand times, and re-estimate the primary results to generate randomization inference p-values. The results, in Appendix Table A9, show the original cluster robust p-values alongside the calculated randomization inference p-values and illustrate very similar levels of statistical significance across this paper's outcomes.

IV Mechanisms

To systematically explore why the treatment improves agricultural outcomes, I refer to Figure 1 where I suggest mechanisms as operating across four margins: 1) The

program causes a decrease in alcohol consumption, 2) lower alcohol consumption leads to positive economic outcomes, 3) the program affects economic outcomes outside the program’s direct effect on alcohol consumption, and 4) the intervention’s direct effect on economic outcomes leads to decreases in alcohol consumption.²⁰ I analyze potential mechanisms on each of these margins in turn below and in Appendix Table A10, providing evidence supporting (or discounting) each.

A Intervention → Lower Alcohol Consumption

The first two mechanisms to consider are related to information received from the program: first, that an individual learns that returns to consumption of alcohol are negative (Mechanism 1.1), and second, that the individual updates their beliefs in their own ability to stop drinking (1.2). Qualitative evidence to support these mechanisms comes from counselor notes. A non-exhaustive list of anonymized quotes from counselor reports in Appendix C demonstrate that indeed, many individuals learned of the negative effects of alcohol, either from the group workshops or the counseling sessions. The quotes include examples of participants relating stories to the counselor of alcohol’s impact on their own lives, such as on health, family relationships, and their finances. The quotes further provide qualitative evidence that the sessions led to personal commitments for sobriety, which suggests individuals have recognized that they have the ability to stop drinking.

The next two potential mechanisms relate to the social aspect of the path towards sobriety: spouse or community members help the individual to reduce/eliminate alcohol consumption (Mechanism 1.3), and that if drinking is social, peer effects reduce the desire to drink (1.4). Evidence for these mechanisms comes from Murphy (2023), which analyzes the social network data collected from participants in this program. The paper shows that among two men i and j who live nearby one another, if both were assigned to the treatment, i is six percentage points more likely to have received help decreasing his alcohol consumption from j compared to two men in the control ($p < 0.05$).

Given the highly religious nature of Kenyan society, another potential mecha-

²⁰I am grateful to an anonymous reviewer for suggesting this framework for analyzing the mechanisms.

nism is that the program led to increases in religiosity of participants, which decreased alcohol consumption (Mechanism 1.5). This would not be a unique causal pathway within the context of sobriety programs, given that many such programs employ religion as a means to achieve sobriety in participants. While the implementing organization was officially secular, there are many instances found in counselor notes of the use of religion and spirituality in their sessions to help control drinking. To gauge whether the intervention caused those assigned to the treatment to become more religious/spiritual, enumerators asked respondents questions from a detailed religious module. The estimates show null effects that are precisely estimated; the treatment had no effect on any religious index (full results are presented in Murphy (2025b)). Thus, a change in religiosity is an unlikely mechanism.

An additional mechanism relates to the pharmacological component of the program (Mechanism 1.6). A staff nurse prescribed withdrawal medication to those in the program experiencing symptoms. The records indicate that, out of the 400 men and women assigned to the treatment, 44 received withdrawal medication. This is not an exhaustive number, as some participants may have experienced minor withdrawal symptoms that were not severe enough to warrant reporting to the nurse.

B Lower Alcohol Consumption → Positive Economic Outcomes

A likely mechanism for this causal pathway is a reallocation of spending from alcohol to productive inputs (Mechanism 2.1), which is evidenced both in counselor notes and in expenditure data. In Table 7, I show estimated ITT effects among men in the medium-run on various spending categories (these expenditure outcomes, with the exception of alcohol expenditures, are not included in the preregistration). As discussed earlier, the estimations show that spending on alcohol was 48 percent less, or approximately 631 KSh, relative to the pure control group over the month prior to the medium-run survey. Furthermore, spending on agriculture and livestock was about 82 percent higher (about 279 KSh) relative to men in the pure control in the month prior to the 2022 survey ($p < 0.1$). Therefore, if men on average are investing about half of the 631 KSh reduction in alcohol spending into agriculture, it would account for the 279 KSh increase in agricultural spending.

However, this consumption expenditure analysis includes some caveats. First,

the agricultural spending category is broad, including expenditures on fertilizer, livestock, and livestock feed. Secondly, it is worth noting that agricultural expenditures are usually lumpy, and the survey question only asks about expenditures in the past month prior to the survey. For these reasons, the changes in spending found in this section cannot be directly compared to the increases in fertilizer usage discussed previously, in which the agricultural module within the survey asked about seasonal application. Finally, most of the expenditure outcomes reported in Table 7 are statistically insignificant when using Romano-Wolf adjusted p-values for multiple hypothesis testing, with the exception being the effect on alcohol spending.

The data on expenditures also provide suggestive evidence for Mechanism 2.2, that improved health leads to better economic outcomes. While the survey did not collect health data *per se*, data exist on whether a respondent paid for any medical services over the prior 30 days. Table 7 shows that men in the treatment spent about 57 percent less on doctor visits and medication over the month prior to the medium-run survey relative to men in the pure control group ($p < 0.05$). However, this result is not robust to p-values adjusted for multiple hypothesis testing.

Unfortunately, there is insufficient evidence in the data to evaluate the following two mechanisms: that participants are more clearheaded and thus able to make smarter choices about inputs (Mechanism 2.3), and that participants are more willing to take risks and thus more willing to use fertilizer (Mechanism 2.4). However, both are intuitively plausible given the overall results of the program.

C Intervention $\rightarrow$ Positive Economic Outcomes

The third margin represents mechanisms originating from the intervention that affect economic outcomes outside of its effect on alcohol consumption. For example, the program may have affected intrahousehold dynamics and improved the relationship between the spouses, which leads an individual to be happier, more willing to work, take risks, and make better business decisions (Mechanism 3.1), or made an individual happier in general (3.2). Evidence for Mechanism 3.1 comes from a companion paper, Castilla, Aqeel and Murphy (2025), which finds that the program decreases intimate partner violence (IPV) in treatment households relative to those in the pure control. The results are particularly evident in the short run, although

effects on economic control are long-lasting. More harmony in the household is likely to make the spouses happier and more willing to work.

In addition, the treatment included group workshops, which brought participants together and gave them a joint objective of decreasing alcohol consumption. Thus, the treatment may have spurred participants towards greater interaction with each other, and given that all households have small farms, led to sharing of information on agriculture and related technologies (Mechanism 3.3). In an accompanying paper and using only the treatment villages in the analysis so as to control for village-level factors, I demonstrate that when two same-gender farmers i and j are both assigned to the treatment, i is five percentage points more likely to say that they received agricultural advice from j in the past six months (nine percentage points more likely if they live nearby one another) compared to two same-gender peers who were both in the spillover control group (Table 2 in Murphy (2023)). Other results show same-gender peers who were both assigned to the treatment are more likely to know each other, indicate they recently worked together with one another, and speak daily with each other than those peers both assigned to the control.

It is also possible that the visits by nurses improved the health of respondents outside of any effect on alcohol consumption (Mechanism 3.4), in contrast to Mechanism 1.6. This may have been due to vitamins given by the nurse to respondents, or because other health issues were concurrently addressed while the nurse was visiting. While there is no direct evidence for this mechanism, I can point to the same empirical results as for Mechanism 1.6, which show that men in the treatment spent less on medical expenses compared to the pure control group.

It may also be that the incentive payments received through the lotteries (discussed in Appendix B) directly affected the agricultural outcomes (Mechanism 3.5). In the program, 164 households received payments (their eligibility was conditional on participation), with a recipient household receiving an average of 1000 KSh. I can thus explore whether treatment households that received any payments also had better agricultural outcomes. The shortcoming to this analysis is that I cannot disentangle the effects of the incentive payments on unobserved factors influencing participation, so these results should be taken with caution. I find, however, that there are no significant relationships between amount of incentive payments

received and any agricultural outcome, although the effects on organic inputs and quantity of hired labor are somewhat large and in a negative direction (results in Appendix Table A11).

D Positive Economic Outcomes → Lower Alcohol Consumption

In analyzing counselor reports, a common recommendation given by counselors is to use agricultural work as a method to control urges to drink. As qualitative evidence for this mechanism, I include example quotes in Appendix C that illustrate both the recommendations by counselors, as well as reports by participants illustrating their compliance with these recommendations. It is possible that these recommendations directly affected agricultural outcomes outside of any effect on alcohol consumption, which in turn caused a decrease in alcohol consumption (Mechanism 4.1). To test this, I run a search through all counselor notes and identify households in which the counselor explicitly records that anything related to work in agriculture was discussed with that household during counseling sessions. In total, 90 of 200 treatment households fall into this category. This may be a significant underestimate, however, as the counselor notes do not record everything discussed in the counseling sessions. I thus conduct new estimations that include an indicator for whether the counselor explicitly recorded that agriculture was discussed with that household in the sessions. This variable has the potential only to be nonzero for those assigned to the treatment. I report the results of this analysis in Appendix Table A12. I find mixed results – estimations show a significant negative relationship between counselors discussing agriculture with the household and endline organic input usage and a positive relationship with endline hired labor. These relationships are potentially endogenous with agricultural outcomes, however, as it may be that counselors discuss farming with more agriculturally inclined households, or alternatively, with those engaged in poor agricultural practices. To further explore this possibility, I analyze whether counselors were more likely to make recommendations to farmers based on baseline versions of the agricultural outcomes. In Appendix Table A13, results suggest a positive relationship between counselor agricultural discussions with baseline harvest value and organic input use, and a negative relationship with quantity of hired labor. Overall, these non-causal and

mixed results do not appear to offer a definitive conclusion on the role of this pathway as a potential mechanism.

V Net Benefits

The net benefits of this program are high. The cost of the intervention itself (excluding survey costs) was 37,327 USD (spent in 2021). Benefits of the program include improved health, higher life expectancy, better family and community relations, and improved agricultural outcomes, among others. A back of the envelope calculation using the results from this study and applying estimates by Wood et al. (2018) that consumption of more than 14 drinks per week decreases life expectancy of men by 1-2 years, I find net benefits of the program ranging between 486,653 and 1,010,632 USD (PPP) from extended years of life alone.²¹ Furthermore, results presented above indicate that harvest values increase by approximately 32 percent (11,570 KSh) among those in the treatment (Table 5). If I annualize the higher agricultural spending of men in the treatment observed on Table 7 as a proxy for costs to the treatment household of the greater agricultural production and I assume uniform effects across all treated, my calculations suggest that the program had agricultural benefits of 14,434 USD (excluding cost of own-household labor) in the one year following the intervention. This would necessitate a value of a statistical life year from improved health of only about 970 USD for the program benefits to exceed costs when including net benefits from the program on agricultural production. Given that Jamison et al. (2013) suggest a value of a statistical life year of 2.3 times the per capita income of a country (which amounts to 13,574 USD (PPP) in the case of Kenya), the benefits significantly exceed the cost.

VI Discussion and Conclusion

I administered an RCT that provided households in rural Kenya with a counseling and pharmacotherapy intervention to reduce heavy alcohol consumption. Those assigned to the treatment decreased unhealthy drinking behaviors relative to those as-

²¹For these net benefit calculations, I use the finding that men assigned to the treatment are 19.3 pp less likely to be a heavy drinker in the medium run (Table 2, bottom panel, Column 15) and a value of statistical life of 2.3 times per-capita income (PPP) (Jamison et al., 2013).

signed to the control. Other notable findings suggest that the intervention increases the use of fertilizer and improves crop production by value.²²

This study took careful steps to minimize attrition, control for confounding factors, and mitigate other potential sources of bias. Project staff asked respondents about both their own and their spouses' alcohol consumption. Furthermore, as an objective measure of alcohol consumption, an enumerator later conducted unannounced spot checks using a breathalyzer. Results using objective breathalyzer data support those using spouse-reported data. Both show significant decreases in the likelihood of drinking and indicate the continued persistence of treatment effects on reductions in alcohol use more than two years later. This is especially notable given that past research finds that up to 80 percent of those with Alcohol Use Disorder who attempt sobriety relapse within twelve months (Moos and Moos, 2006). This study illustrates that a well-designed counseling program coupled with pharmacotherapy can minimize relapse in participants over extended periods.

While it is difficult to precisely disentangle the causal pathways linking the intervention to various outcomes, I systematically explore many potential mechanisms including a reallocation of funds from alcohol to agricultural inputs, the use of medication to reduce withdrawal symptoms, improved health, and an increase in information and labor sharing between peers. A confluence of various mechanisms likely contributed to the program's success. Therefore, to replicate this program at a larger scale while achieving similar results, a key consideration would be to ensure the potential mechanisms identified above have the opportunity to operate. This means that in addition to individual counseling, the program should include both group workshops to create/strengthen bonds across individuals in same-village

²²Indeed, the effect on fertilizer usage in this study (a 31 percent increase - equivalent to a 0.47 percent increase in fertilizer use per USD of reduction in male annualized alcohol expenditures) is nearly as large as other studies that had the explicit goal of increasing fertilizer use. As a comparison, also in western Kenya, Duflo, Kremer and Robinson (2011) find that offers to purchase fertilizer for the next cropping season shortly after harvest coupled with free delivery increase fertilizer usage by 47 to 70 percent. Karlan et al. (2011) find a loan indemnity program in Ghana increases chemical input usage by 35 percent. In a longer-run study, Carter, Laajaj and Yang (2021) in Mozambique find that a program that provided a 73 percent subsidy for a package of inputs that included chemical fertilizer led to a 118 percent increase in fertilizer applied on maize in the year of the program (a 1.38 percent increase in fertilizer use per USD of voucher) relative to control group farmers, and a 35 percent increase over the two subsequent years after the subsidy had expired.

households and spousal counseling to improve relationships within the household. To maintain high levels of participation, counseling sessions should be conducted at the residence, which minimizes opportunity costs of participation. Furthermore, the coupling of counseling and pharmacotherapy can address both motivational and biochemical obstacles to sobriety.

One limitation of this study is its inclusion only of cohabiting spouses; it excludes those who have spouses that are deceased or who have migrated to cities for work. Two factors mitigate concerns about this selection issue. First, the treatment program relies on the effectiveness of Motivational Enhancement Therapy in a family systems approach, which builds trust and accountability between spouses for alcohol rehabilitation and to prevent relapse. Second, most single-adult households in western Kenya are female-led since men have a lower life expectancy and are more likely to migrate for work. Because women on average drink less much than men, it is arguable that single-adult households excluded from my study do not contain the individuals of greatest concern.

Though the sample is limited to rural western Kenya, the effects of this program are likely externally valid and transferable to other developing country contexts. Most of SSA faces similar problems of high levels of alcohol abuse in rural areas, and most of that alcohol is informally produced and unregulated (WHO, 2024). Addressing abuse of informal alcohol from supply-side policies, such as through taxes or prohibition, is challenging and often ineffective. Given high net-benefits and potential to scale, this study demonstrates that relatively inexpensive demand-side interventions involving workshops, counseling, and nurse visits has potential for large and significant impacts on alcohol consumption and economic outcomes.

More broadly, the results of this study add to the growing body of evidence evaluating whether cognitive behavioral therapy (CBT) can address constraints to economic development. These programs may be especially effective in rural areas of both developed and less developed countries where formal health services are often limited. Given these programs are relatively inexpensive and net benefits potentially large, future research and policy efforts should explore the feasibility of expanding counselor-led CBT programs as a tool for poverty reduction.

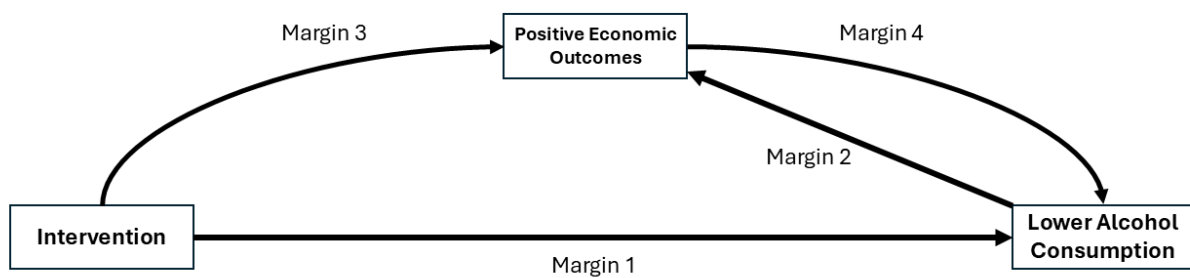


FIGURE 1: Margins of Causal Mechanisms

Notes: Margins 1, 2, 3, and 4 correspond to the same numbered margins on Appendix Table A10.

TABLE 1
SUMMARY STATISTICS OF INDIVIDUAL-LEVEL OUTCOME VARIABLES

Variable	Pooled Sample	Men	Women
Drank in past week (binary)	0.47 (0.50)	0.69 (0.46)	0.26 (0.44)
Quantity consumed in past week (US Std. drinks)	10.36 (18.48)	18.20 (22.80)	2.54 (6.47)
Heavy drinker (binary) [†]	0.26 (0.44)	0.40 (0.49)	0.11 (0.31)
Maximum drinks consumed in one day in past week	3.07 (4.61)	5.28 (5.39)	0.87 (1.96)
Spouse reported: drank in past week (binary)	0.44 (0.50)	0.75 (0.43)	0.12 (0.32)
Spouse reported: quantity consumed in past week (US Std. drinks)	11.84 (19.36)	22.33 (22.33)	1.34 (5.58)
Spouse reported: heavy drinker (binary) [†]	0.30 (0.46)	0.55 (0.50)	0.06 (0.23)
Spouse reported: max alcohol consumed in one day in past 7 days	2.79 (4.45)	5.14 (5.13)	0.45 (1.50)
Any wage work for others in past month (binary)	0.66 (0.48)	0.82 (0.38)	0.49 (0.50)
Any non-farm wage work for others in past month (binary)	0.40 (0.49)	0.63 (0.48)	0.18 (0.38)
Total earnings from labor in past month [‡]	6,830.48 (10,794.06)	11,180.65 (13,174.97)	2,489.98 (4,693.01)
Hours of labor yesterday own farm	1.98 (2.37)	2.37 (2.63)	1.58 (2.02)
Hours of self employment yesterday	1.84 (3.11)	2.36 (3.64)	1.33 (2.38)
Hours of wage labor yesterday	1.81 (3.29)	2.65 (3.83)	0.98 (2.36)
Observations	899	449	450

Notes: Table shows mean and standard deviations for data collected in the medium run. Alcohol measures are US standard drink equivalents; one standard serving of the local liquor chang'aa is equivalent to approximately 5.75 US standard drinks. [†] A heavy drinker is defined as at least 15 (8) standard drinks consumed in the past week for a man (woman). [‡] I calculate "Total earnings from labor" as earnings in Kenyan Shillings from self employment (calculated after Agness et al. (2025)) and actual market wages over the past 30 days.

TABLE 2
SPOUSE-REPORTED ALCOHOL USE PAST WEEK

	Ext. margin: Short run			Ext. margin: Medium run		
	(1)	(2)	(3)	(4)	(5)	(6)
Assigned to treatment	-0.177 (0.048) [0.002]	-0.303 (0.058) [0.000]	-0.043 (0.050) [0.444]	-0.122 (0.037) [0.004]	-0.190 (0.043) [0.001]	-0.053 (0.051) [0.351]
Spillover control	-0.090 (0.048) [0.115]	-0.111 (0.058) [0.094]	-0.068 (0.053) [0.243]	-0.054 (0.039) [0.195]	-0.117 (0.052) [0.026]	-0.002 (0.053) [0.965]
Constant	0.474 (0.043) [0.000]	0.782 (0.046) [0.000]	0.166 (0.044) [0.000]	0.500 (0.027) [0.000]	0.859 (0.016) [0.000]	0.140 (0.043) [0.000]
Sample	M,W	M	W	M,W	M	W
Pure control mean	0.474	0.779	0.169	0.500	0.856	0.144
b[Treat]=b[SC] (p-val)	0.005	0.000	0.414	0.018	0.145	0.187
Treatment lower bound	-0.187	-0.321	-0.046	-0.125	-0.197	-0.053
Treatment upper bound	-0.164	-0.298	-0.023	-0.121	-0.189	-0.053
Treat: MHT (p-val) [†]	0.051	0.009	0.642	0.049	0.002	0.632
Observations	982	491	491	874	437	437
	Heavy drinker: Short run			Heavy drinker: Medium run		
	(1)	(2)	(3)	(4)	(5)	(6)
Assigned to treatment	-0.145 (0.030) [0.000]	-0.274 (0.054) [0.000]	-0.021 (0.020) [0.345]	-0.112 (0.033) [0.005]	-0.193 (0.057) [0.005]	-0.039 (0.036) [0.399]
Spillover control	-0.056 (0.029) [0.090]	-0.062 (0.059) [0.314]	-0.040 (0.017) [0.043]	-0.056 (0.038) [0.151]	-0.126 (0.060) [0.037]	0.012 (0.040) [0.819]
Constant	0.312 (0.020) [0.000]	0.567 (0.038) [0.000]	0.057 (0.015) [0.002]	0.366 (0.021) [0.000]	0.665 (0.029) [0.000]	0.068 (0.034) [0.002]
Sample	M,W	M	W	M,W	M	W
Pure control mean	0.312	0.565	0.058	0.367	0.662	0.072
b[Treat]=b[SC] (p-val)	0.001	0.000	0.233	0.091	0.282	0.050
Treatment lower bound	-0.152	-0.287	-0.022	-0.114	-0.197	-0.039
Treatment upper bound	-0.126	-0.262	0.005	-0.111	-0.191	-0.039
Treat: MHT (p-val) [†]	0.000	0.000	0.687	0.022	0.021	0.459
Observations	982	491	491	874	437	437

Notes: ANHECOVA estimations after Ye et al. (2023), which are calculated using linear probability models. Dependent variable for extensive margin (top panel) is whether any alcohol was consumed in the week prior to the survey (spouse-reported). “Baseline: Alcohol past week (binary)” measures whether the respondent drank in the past week in the baseline survey (self-reported). Heavy drinker (bottom panel) defined as consumption of at least 15 (8) US standard drink equivalents per week for men (women). Sample refers to pooled gender (M,W), Men (M), or Women (W). Demeaned variables included but not shown: control for male drinker (and its interactions with treatment group) in pooled gender estimates, baseline alcohol consumption and its interactions with treatment group. “b[Treat]=b[Sp. control]” reports the p-value for a Wald test on whether the estimated coefficient of the treatment is statistically different from the estimated coefficient of the spillover control. I calculate upper and lower bounds following Lee (2009) and McKenzie (2017). Medium-run estimations only include sample that was nonattrited in short-run (i.e., excludes those found and surveyed in 2022 but not 2021). [†]I calculate multiple hypothesis test p-values here following List, Shaikh and Xu (2019) and Steinmayr (2020), which control for the familywise error rate using bootstrapped resampling and allows for p-values to be correlated. Heteroskedasticity-robust standard errors are clustered at the village level, and wild bootstrap p-values shown in brackets below estimated coefficients and standard errors. I present alternative ANOVA and ANCOVA estimations in Appendix Table A6 to demonstrate robustness.

TABLE 3
SPOUSE-REPORTED INTENSIVE MARGIN EFFECTS AMONG MEN

	Int. Margin: Short run			Int. Margin: Medium run		
	(1)	(2)	(3)	(4)	(5)	(6)
		Lower	Upper		Lower	Upper
Assigned to treatment	-1.193 (0.192)	-1.533 (0.199)	-0.313 (0.196)	-0.583 (0.116)	-0.772 (0.115)	-0.276 (0.117)
Constant	3.584 (0.104)	3.924 (0.117)	2.705 (0.113)	3.415 (0.051)	3.604 (0.049)	3.108 (0.052)
Pure control mean	37.545	37.545	37.545	31.178	31.178	31.178
Treat: MHT (p-val)‡	0.011			0.000		
Observations	337	293	293	305	281	281
	Max drinks: Short run			Max drinks: Medium run		
	(1)	(2)	(3)	(4)	(5)	(6)
		Lower	Upper		Lower	Upper
Assigned to treatment	-1.106 (0.161)	-1.439 (0.171)	-0.391 (0.180)	-0.449 (0.124)	-0.637 (0.127)	-0.072 (0.117)
Constant	2.221 (0.091)	2.554 (0.107)	1.506 (0.122)	1.903 (0.056)	2.091 (0.060)	1.526 (0.035)
Pure control mean	9.562	9.562	9.562	6.791	6.791	6.791
Treat: MHT (p-val)‡	0.004			0.010		
Observations	337	293	293	305	281	281

Notes: Table shows estimates of β using a Poisson QMLE model (Correia, Guimarães and Zylkin, 2020). Coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Intensive margin dependent variable: spouse-reported alcohol consumption of individual in past seven days - US standard drink equivalent. In bottom panel outcome variable: maximum drinks in one day in the past week, which represents the spouse-reported greatest number of drinks consumed within one day in the past week. †I calculate multiple hypothesis test p-values here following List, Shaikh and Xu (2019) and Steinmayr (2020), which control for the familywise error rate using bootstrapped resampling and allows for p-values to be correlated. MHT p-values are calculated here using linear estimations. I calculate intensive margin bounds following Lee (2009) and Chen and Roth (2024), trimming the pure control group at the top and bottom of the distribution to account for the share of “always-takers” in the sample. All estimates presented on this table exclude the spillover control sample and compare the treatment directly to the pure control. Results for women cannot be calculated given that most women consume no alcohol. Heteroskedasticity-robust standard errors are clustered at the village level.

TABLE 4
BREATHALYZER RESULTS

	Any alcohol detected (binary)		
	(1)	(2)	(3)
Assigned to treatment	-0.147 (0.034) [0.000]	-0.136 (0.035) [0.000]	-0.143 (0.038) [0.000]
Spillover control	-0.109 (0.057) [0.086]	-0.085 (0.056) [0.137]	-0.082 (0.059) [0.174]
Constant	0.361 (0.028) [0.000]	0.350 (0.028) [0.000]	0.359 (0.032) [0.000]
Estimation†	ANA	ACV	AHV
Pure control mean	0.361	0.361	0.361
Treatment lower bound	-0.155	-0.146	-0.153
Treatment upper bound	-0.132	-0.120	-0.126
Treat: MHT (p-val)‡	0.006	0.022	0.071
Observations	423	423	423

Notes: †ANOVA (ANA), ANCOVA (ACV), and ANHECOVA (AHV) estimations after Ye et al. (2023) are calculated using linear probability models. Breathalyzer only conducted among men in the sample. Dependent variable is whether $BAC > 0.01$, i.e., a positive test for alcohol consumption outside the margin of error. Demeaned variables included but not shown: baseline alcohol consumption, age, time of test, whether test was done on a weekend in ANCOVA, ANHECOVA and their interactions with treatment group in the latter. Upper and lower bounds are calculated following Lee (2009) and McKenzie (2017). Those who refused the breathalyzer (9 individuals) are considered to have drunk alcohol. ‡I calculate multiple hypothesis test p-values following List, Shaikh and Xu (2019) and Steinmayr (2020), which control for the familywise error rate using bootstrapped resampling and allows for p-values to be correlated. These tests were not included in the initial project plan and therefore this outcome is not included in the preregistration. Heteroskedasticity-robust standard errors are clustered at the village level and wild bootstrap p-values are shown in brackets below estimated coefficients and standard errors.

TABLE 5
AGRICULTURAL OUTCOMES

	Harvest Value 1000s KSh			Inorganic Fertilizer Kg			Organic Inputs Kg			Hired Labor (Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Assigned to treatment	0.310 (0.234)	0.302 (0.091)	0.278 (0.098)	0.218 (0.206)	0.221 (0.089)	0.270 (0.099)	-0.335 (0.324)	-0.240 (0.341)	-0.394 (0.281)	0.209 (0.195)	0.194 (0.181)	0.424 (0.183)
Spillover control	0.129 (0.199)	0.169 (0.063)	0.086 (0.072)	-0.001 (0.186)	-0.039 (0.077)	0.024 (0.089)	-0.467 (0.312)	-0.336 (0.331)	-0.439 (0.272)	0.128 (0.193)	0.073 (0.159)	0.156 (0.185)
Constant	3.584 (0.162)	3.381 (0.048)	3.410 (0.059)	4.138 (0.134)	3.936 (0.064)	3.875 (0.068)	4.785 (0.218)	4.640 (0.215)	4.710 (0.170)	1.907 (0.129)	1.696 (0.111)	1.486 (0.138)
Estimation [†]	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV
b[Treat]=b[Sp. control] (p-val)	0.038	0.081	0.005	0.053	0.006	0.017	0.729	0.804	0.901	0.676	0.589	0.142
Pure control mean	36.171	36.171	36.171	63.282	63.282	63.282	118.915	118.915	118.915	6.944	6.944	6.944
Lee lower bound	0.175	0.205	0.239	0.113	0.188	0.257	-0.640	-0.568	-0.735	0.091	0.095	0.372
Lee upper bound	0.322	0.306	0.287	0.236	0.233	0.286	-0.317	-0.222	-0.379	0.227	0.202	0.439
Treat: MHT (p-val) [‡]	0.313	0.053	0.046	0.563	0.102	0.065	0.563	0.561	0.216	0.563	0.561	0.086
Observations	448	448	448	448	448	448	448	448	448	448	448	448

Notes: [†]ANOVA (ANA), ANCOVA (ACV), and ANHECOVA (AHV) estimations after Ye et al. (2023) are calculated using Poisson QMLE (Correia, Guimarães and Zylkin, 2020). Coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Outcomes are annual (two season) totals. Agricultural price data in 2022 is deflated to 2020 levels using Kenya Agricultural PPI (FAO, 2023). Demeaned variables included in ANCOVA but not shown: baseline of outcome, an indicator for whether farmers had fully completed the 2022 long rains harvest (see text for additional information on this variable), plot area and its square. These variables are interacted with treatment group assignment in ANHECOVA estimations. [‡]I calculate multiple hypothesis test p-values here following Romano and Wolf (2016) and Clarke, Romano and Wolf (2020), which control for the familywise error rate and uses bootstrapped resampling to allow for dependence among the p-values. I calculate upper and lower bounds following Lee (2009) and McKenzie (2017). I include baseline regressions of outcomes in Appendix Table A7, which illustrates all outcomes are balanced at baseline with the exception of organic input use, which is significantly lower among those in treatment at baseline. Heteroskedasticity-robust standard errors are clustered at the village level.

TABLE 6
HOURS WORKED YESTERDAY & WAGES PAST MONTH

	Hours worked own farm			Hours worked self employ			Hours worked for wages		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Assigned to treatment	0.043 (0.120)	-0.053 (0.183)	0.143 (0.180)	0.198 (0.155)	0.228 (0.195)	0.193 (0.248)	-0.316 (0.168)	-0.166 (0.139)	-0.401 (0.289)
Spillover control	0.034 (0.105)	-0.014 (0.168)	0.088 (0.180)	0.233 (0.161)	0.260 (0.191)	0.107 (0.246)	-0.427 (0.179)	-0.412 (0.156)	-0.343 (0.292)
Constant	0.630 (0.082)	0.879 (0.154)	0.370 (0.148)	0.352 (0.115)	0.589 (0.118)	0.070 (0.194)	0.690 (0.101)	1.116 (0.071)	0.141 (0.196)
Sample	M,W	M	W	M,W	M	W	M,W	M	W
Pure control mean	1.946	2.443	2.443	1.586	1.946	1.225	2.189	3.046	1.332
Treat: MHT (p-val)‡	0.993	0.993	0.946	0.859	0.900	0.972	0.527	0.941	0.871
Observations	885	442	443	885	442	443	885	442	443
	Any wage earnings			Any non-farm wage earnings			Total labor earnings†		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Assigned to treatment	-0.057 (0.045)	-0.002 (0.042)	-0.116 (0.063)	-0.010 (0.034)	0.003 (0.049)	-0.029 (0.043)	-0.164 (0.130)	-0.098 (0.112)	-0.269 (0.166)
Spillover control	-0.081 (0.046)	-0.038 (0.048)	-0.130 (0.062)	-0.008 (0.035)	-0.013 (0.045)	-0.010 (0.055)	-0.063 (0.169)	-0.157 (0.118)	0.022 (0.232)
Constant	0.700 (0.036)	0.829 (0.032)	0.568 (0.050)	0.411 (0.023)	0.631 (0.032)	0.189 (0.036)	1.734 (0.100)	2.482 (0.068)	0.962 (0.139)
Sample	M,W	M	W	M,W	M	W	M,W	M	W
Pure control mean	0.696	0.829	0.564	0.696	0.636	0.186	7.344	12.003	2.684
Treat: MHT (p-val)‡	0.782	0.997	0.518	0.993	0.997	0.972	0.826	0.972	0.782
Observations	885	442	443	885	442	443	885	442	443

Notes: ANHECOVA estimations after Ye et al. (2023). Top panel estimations and estimates for total labor earnings in bottom panel use Poisson QMLE (Correia, Guimarães and Zylkin, 2020). These coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Sample refers to pooled gender (M,W), Men (M), or Women (W). Demeaned variables included in ANHECOVA but not shown: an indicator for male gender (in pooled sample), baseline alcohol consumption, baseline hours worked, and the interactions between these variables and treatment assignment. †I calculate “Total earnings from labor” as earnings from self employment (calculated after Agness et al. (2025)) and actual market wages over the past 30 days in thousands of Kenyan Shillings. “Hours worked” outcomes use data collection adapted from Field et al. (2023), in which respondents were asked to divide up their last 24 hours into nine categories, including sleeping, working on own farm, self employment, unpaid household work, etc. ‡I calculate multiple hypothesis test p-values here following Romano and Wolf (2016) and Clarke, Romano and Wolf (2020), which control for the familywise error rate and uses bootstrapped resampling to allow for dependence among the p-values. Heteroskedasticity-robust standard errors are clustered at the village level. I include ANOVA estimates for these outcomes in Appendix Table A8.

TABLE 7
EXPENDITURES IN PAST MONTH AMONG MEN (KENYAN SHILLINGS)

Outcome Variable	Treat Coefficient	Model P-val	Romano-Wolf P-val	Pure Control Mean (KSh past month)
Total spending	-0.062	0.605	0.990	17,472.4
Total spending excluding food and alcohol	-0.054	0.730	0.995	11,757.6
Amount saved	0.173	0.462	0.978	897.9
Money received as gifts	0.024	0.942	0.995	324.1
Money given as gifts	0.025	0.918	0.995	624.7
Alcohol spending	-0.653	0.000	0.035	1,314.3
Food spending	0.045	0.656	0.995	4,400.4
Cigarette spending	-0.522	0.246	0.952	90.6
Medical spending	-0.840	0.012	0.365	1,207.6
School spending	-0.068	0.755	0.995	5,758.9
Cooking fuel spending	0.192	0.519	0.986	196.8
Wedding/funeral spending	0.277	0.232	0.952	615.1
Hygiene spending	-0.175	0.304	0.955	357.1
Home repairs/renovations spending	0.236	0.698	0.995	321.5
Electricity spending	0.195	0.393	0.971	221.8
Phone spending	-0.184	0.228	0.952	573.1
Agricultural and livestock spending	0.599	0.092	0.809	339.7
<i>N</i>	442			

Notes: ANOVA estimations. Calculated using Poisson QMLE (Correia, Guimarães and Zylkin, 2020). Coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Data from 16-month post intervention survey in which enumerators asked respondents about consumption expenditures over the past month. For consistency, alcohol and food data, which were collected as weekly data in the survey, are multiplied by four to be equivalent to the monthly data. I calculate multiple hypothesis test p-values here following Romano and Wolf (2016) and Clarke, Romano and Wolf (2020), which control for the familywise error rate and uses bootstrapped resampling to allow for dependence among the p-values. Heteroskedasticity-robust standard errors are clustered at the village level.

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APPENDICES

For Online Publication

A Alcohol, Labor, and Agriculture

Murphy, 2025

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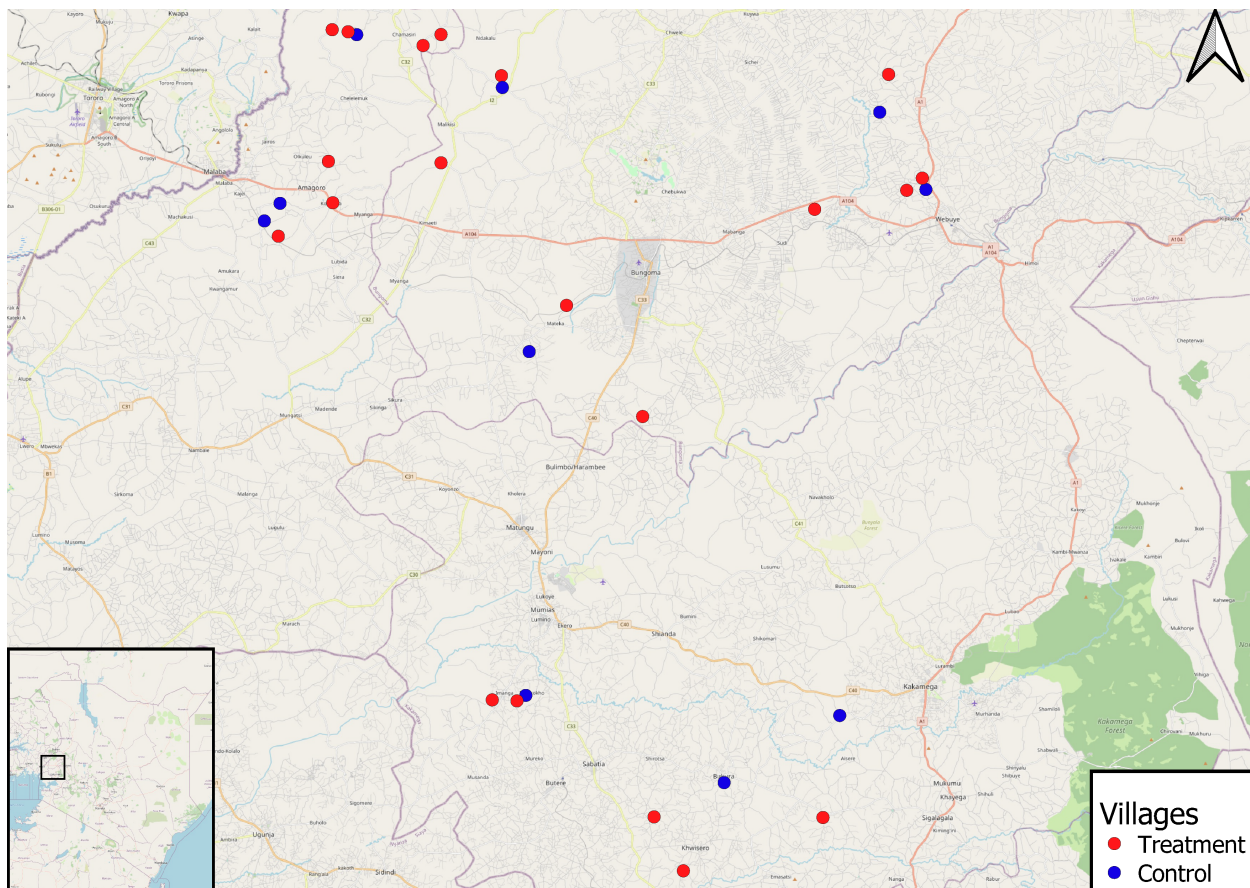


FIGURE A1: Map of Project Area

Notes: Map indicates village locations, with locations differentiated by whether they were treatment or control.

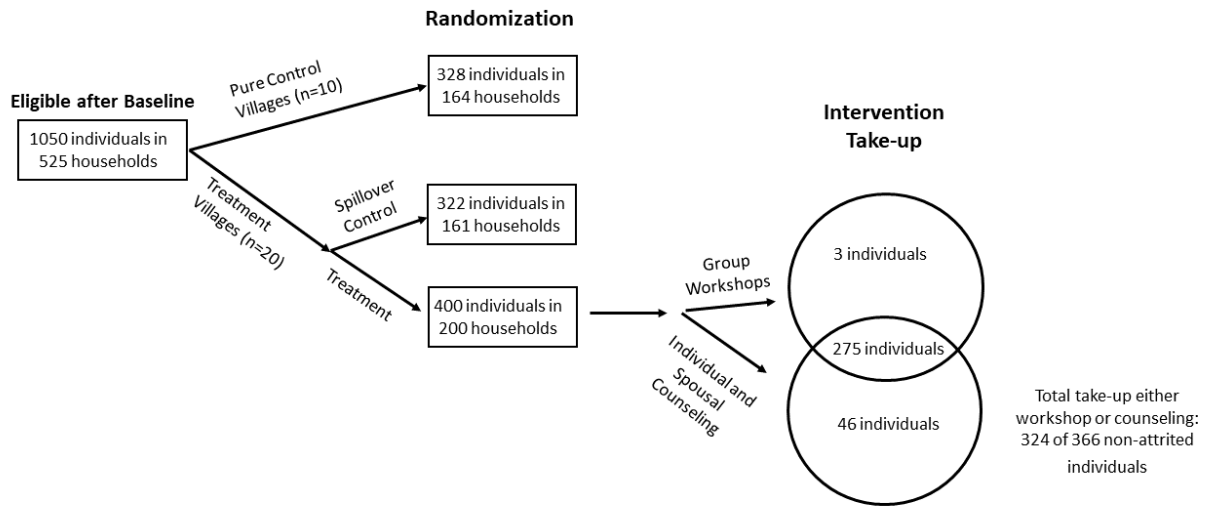


FIGURE A2: Illustration of Randomization

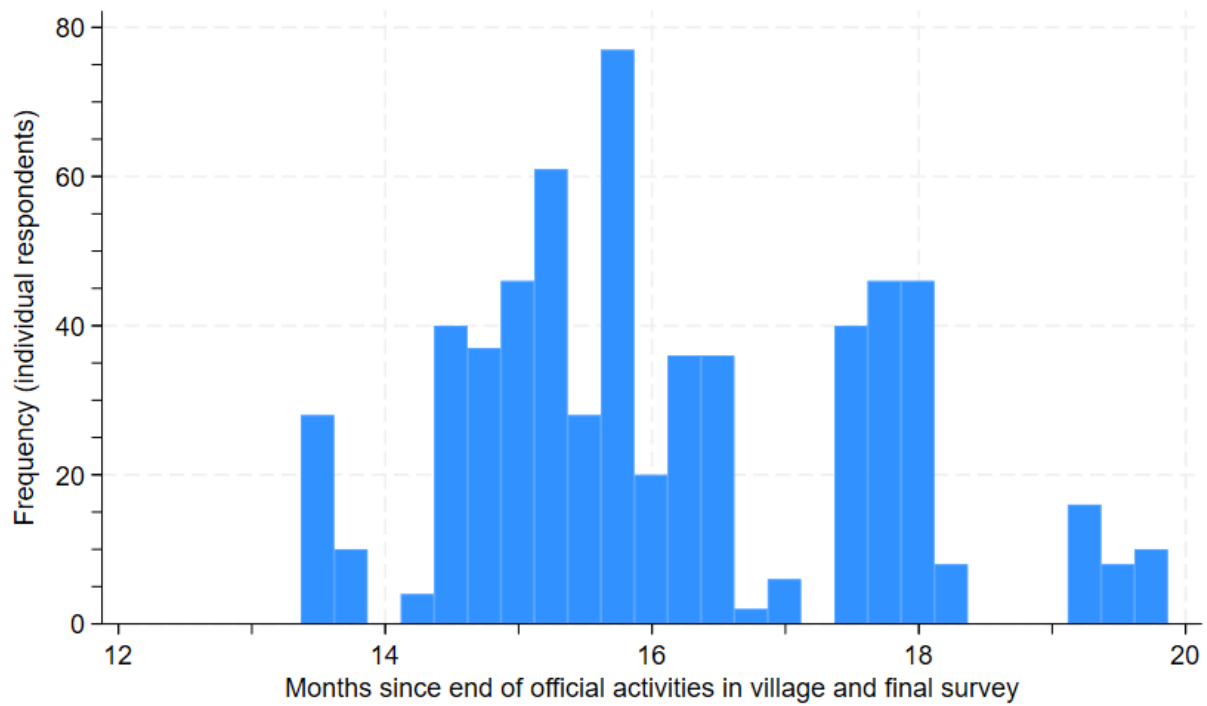


FIGURE A3: Time between End of Intervention and Medium-run Endline Survey

Notes: Histogram includes project participants in treatment villages (i.e., in treatment and spillover control groups).

TABLE A1
DIRECT BALANCE TESTS

	Number drinks past week			Age			Housework hours		
	(1) BL	(2) SR	(3) MR	(4) BL	(5) SR	(6) MR	(7) BL	(8) SR	(9) MR
Assigned to treatment	0.017 (0.135)	0.042 (0.138)	0.010 (0.142)	-0.984 (0.679)	-0.822 (0.709)	-1.161 (0.730)	0.438 (0.898)	0.497 (0.938)	0.824 (0.973)
Spillover control	-0.140 (0.142)	-0.116 (0.144)	-0.187 (0.149)	-0.936 (0.715)	-1.049 (0.739)	-1.209 (0.768)	0.396 (0.946)	0.417 (0.977)	0.155 (1.024)
Constant	1.445 (0.100)	1.409 (0.102)	1.427 (0.105)	37.909 (0.504)	38.029 (0.522)	38.441 (0.541)	18.349 (0.666)	18.232 (0.691)	18.254 (0.721)
Observations	1050	982	899	1050	982	899	1050	982	899
	Male			Own farm hours			Non ag hours		
	(1) BL	(2) SR	(3) MR	(4) BL	(5) SR	(6) MR	(7) BL	(8) SR	(9) MR
Assigned to treatment	0.000 (0.037)	0.000 (0.039)	-0.001 (0.040)	0.858 (1.271)	0.541 (1.315)	1.354 (1.343)	0.486 (1.656)	0.185 (1.712)	1.270 (1.781)
Spillover control	0.000 (0.039)	0.000 (0.040)	0.005 (0.042)	-0.554 (1.339)	-0.325 (1.371)	0.352 (1.413)	1.645 (1.744)	1.737 (1.784)	2.872 (1.875)
Constant	0.500 (0.028)	0.500 (0.029)	0.498 (0.030)	26.457 (0.942)	26.484 (0.969)	26.210 (0.995)	18.134 (1.228)	17.938 (1.262)	17.320 (1.320)
Observations	1050	982	899	1050	982	899	1050	982	899

Notes: Direct balance tests on baseline variables. Results indicate simple difference in means with no controls or adjustments. Column 1 for each dependent variable (“BL”) is the full baseline sample, Column 2 (“SR”) is the sample in the short-run survey round, and Column 3 (“MR”) is the sample in the medium-run survey round. “Number of drinks past week” indicates the number of drinks consumed the last time the respondent drank in the past week at baseline. Hours variables are hours worked in the past week in the following capacities at baseline: hours worked doing housework (“Housework hours”), hours worked on own farm (“Own farm hours”), hours worked in non-agricultural wage labor (“Non ag hours”).

TABLE A2
NUMBER OF COUNSELING SESSIONS

Number of counseling sessions attended	No.	Row %
0	79	19.75
1	7	1.75
2	17	4.25
3	22	5.50
4	12	3.00
5	20	5.00
6	19	4.75
7	39	9.75
8	166	41.50
9	6	1.50
10	7	1.75
11	5	1.25
12	1	0.25
Total	400	100.00

Notes: Table reports number of therapy sessions participated in by individuals in the treatment group. Included in the totals are individuals who eventually attrited and were not resurveyed in future data collection rounds. The intervention goal was eight sessions per household. Individuals receiving more than eight sessions was due to efforts to revisit households that had only one spouse in attendance during some sessions.

TABLE A3
SURVEY MONTH AND TREATMENT STATUS

Month (2022)	May	June	July	August	Sept.	Oct.	Total
Treatment	47	71	76	26	96	26	342
Spillover Control	32	65	69	26	72	12	276
Pure Control	16	52	45	92	42	34	281
Total	95	188	190	144	210	72	899

Notes: Table shows distribution of when a non-attrited individual was surveyed by month (2022) in the medium-run survey. Enumerators surveyed villages and households in a random order.

TABLE A4
ATTRITION: OLS ESTIMATES

	Tracked between baseline and short-run survey			Tracked between baseline and medium-run			Tracked between baseline and breathalyzer test (Men)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Assigned to treatment (Treat)	-0.024 (0.023)	-0.021 (0.023)	-0.020 (0.024)	-0.002 (0.040)	0.003 (0.042)	0.006 (0.038)	-0.016 (0.043)	-0.010 (0.043)	0.028 (0.073)
Spillover control (SC)	[0.324]	[0.371]	[0.413]	[0.963]	[0.954]	[0.881]	[0.724]	[0.840]	[0.736]
	0.017 (0.023)	0.019 (0.022)	0.019 (0.023)	0.000 (0.040)	0.003 (0.041)	0.006 (0.038)	0.003 (0.040)	0.006 (0.043)	0.050 (0.069)
Baseline: Age	[0.470]	[0.413]	[0.447]	[0.993]	[0.939]	[0.896]	[0.945]	[0.901]	[0.544]
		0.002 (0.001)	0.002 (0.002)		0.005 (0.001)	0.007 (0.002)		0.003 (0.002)	0.003 (0.005)
Male		[0.049]	[0.425]		[0.000]	[0.020]		[0.182]	[0.571]
		-0.000 (0.017)	0.004 (0.042)		0.000 (0.027)	-0.030 (0.053)			
Baseline: Alcoholic drinks		[0.985]	[0.973]		[0.997]	[0.613]			
		-0.008 (0.006)	-0.014 (0.016)		-0.015 (0.009)	-0.009 (0.020)		-0.016 (0.010)	0.008 (0.022)
Baseline: Hours worked own farm		[0.222]	[0.545]		[0.129]	[0.805]		[0.127]	[0.746]
		-0.000 (0.001)	-0.000 (0.001)		0.000 (0.001)	-0.002 (0.001)		0.002 (0.001)	-0.000 (0.002)
Baseline: Hours non-farm wage work		[0.634]	[0.862]		[0.606]	[0.124]		[0.091]	[0.805]
		-0.001 (0.000)	-0.001 (0.000)		-0.000 (0.001)	-0.002 (0.001)		-0.000 (0.001)	-0.001 (0.001)
Baseline: Hours engaged in housework		[0.089]	[0.266]		[0.636]	[0.099]		[0.634]	[0.693]
		-0.001 (0.001)	-0.001 (0.001)		0.000 (0.001)	-0.001 (0.002)		-0.001 (0.002)	-0.003 (0.005)
Treat × Baseline: Age		[0.216]	[0.394]		[0.981]	[0.511]		[0.570]	[0.512]
			0.002 (0.003)			-0.002 (0.004)			0.001 (0.006)
SC × Baseline: Age			[0.511]			[0.543]			[0.819]
			-0.002 (0.002)			-0.004 (0.003)			-0.000 (0.006)
Treat × Male			[0.481]			[0.251]			[0.955]
			-0.008 (0.050)			0.048 (0.068)			
SC × Male			[0.878]			[0.523]			
			-0.004 (0.045)			0.028 (0.061)			
Treat × Baseline: Alcoholic drinks			[0.947]			[0.686]			
			0.008 (0.019)			-0.001 (0.024)			-0.034 (0.027)
SC × Baseline: Alcoholic drinks			[0.679]			[0.957]			[0.253]
			0.007 (0.017)			-0.022 (0.025)			-0.038 (0.025)
Treat × Baseline: Hours worked own farm			[0.712]			[0.407]			[0.167]
			-0.001 (0.001)			0.002 (0.002)			0.005 (0.002)
SC × Baseline: Hours worked own farm			[0.428]			[0.235]			[0.023]
			0.001 (0.001)			0.004 (0.002)			-0.000 (0.003)
Treat × Baseline: Hours non-farm wage work			[0.393]			[0.031]			[0.987]
			-0.001 (0.001)			0.002 (0.001)			0.000 (0.002)
SC × Baseline: Hours non-farm wage work			[0.491]			[0.104]			[0.917]
			0.000 (0.001)			0.003 (0.001)			0.001 (0.002)
Treat × Baseline: Hours engaged in housework			[0.600]			[0.064]			[0.581]
			0.001 (0.002)			0.004 (0.002)			0.001 (0.006)
SC × Baseline: Hours engaged in housework			[0.674]			[0.172]			[0.827]
			0.000 (0.001)			-0.001 (0.002)			0.003 (0.006)
Constant	0.939 (0.016)	0.937 (0.016)	0.938 (0.016)	0.857 (0.031)	0.854 (0.033)	0.850 (0.029)	0.811 (0.031)	0.807 (0.040)	0.772 (0.061)
	[0.000]	[0.001]	[0.000]	[0.000]	[0.001]	[0.000]	[0.001]	[0.000]	[0.000]
Mean of dep. variable		0.935			0.856			0.806	
Observations	1050	1050	1050	1050	1050	1050	525	525	525

Notes: Treat=Assigned to treatment. SC=Spillover control. The reference group is those in the pure control. Breathalyzer tests only conducted among men in the sample. Heteroskedasticity-robust standard errors are clustered at the village level and wild bootstrap p-values are shown in brackets below estimated coefficients and standard errors.

TABLE A5
HOUSEHOLD-LEVEL ATTRITION: OLS ESTIMATES

	Tracked between baseline and short-run survey			Tracked between baseline and medium-run		
	(1)	(2)	(3)	(4)	(5)	(6)
Assigned to treatment	-0.024 (0.023) [0.324]	-0.020 (0.023) [0.400]	-0.019 (0.024) [0.457]	0.014 (0.042) [0.744]	0.019 (0.044) [0.674]	0.022 (0.039) [0.575]
Spillover control	0.017 (0.023) [0.470]	0.020 (0.022) [0.385]	0.020 (0.024) [0.435]	-0.002 (0.041) [0.962]	0.000 (0.043) [0.996]	0.002 (0.039) [0.949]
Baseline: Age		0.001 (0.001) [0.040]	0.001 (0.001) [0.412]		0.003 (0.001) [0.000]	0.004 (0.001) [0.007]
Baseline: Alcoholic drinks past week§		-0.007 (0.005) [0.218]	-0.010 (0.012) [0.549]		-0.012 (0.008) [0.136]	-0.010 (0.015) [0.720]
Baseline: Hours worked own farm		-0.000 (0.000) [0.660]	-0.000 (0.001) [0.873]		0.000 (0.001) [0.456]	-0.001 (0.001) [0.271]
Baseline: Hours non-farm wage work		-0.001 (0.000) [0.083]	-0.001 (0.000) [0.298]		-0.000 (0.000) [0.522]	-0.002 (0.001) [0.079]
Baseline: Hours engaged in housework		-0.001 (0.001) [0.229]	-0.001 (0.001) [0.371]		0.000 (0.001) [0.674]	0.000 (0.001) [0.924]
Treat × Baseline: Age			0.001 (0.002) [0.615]			-0.002 (0.002) [0.380]
Treat × Baseline: Alcoholic drinks past week§			0.006 (0.015) [0.747]			0.006 (0.019) [0.770]
Treat × Baseline: Hours worked own farm			-0.001 (0.001) [0.486]			0.001 (0.001) [0.223]
Treat × Baseline: Hours non-farm wage work			-0.000 (0.001) [0.594]			0.002 (0.001) [0.109]
Treat × Baseline: Hours engaged in housework			0.000 (0.002) [0.814]			0.003 (0.002) [0.152]
SC × Baseline: Age			-0.001 (0.001) [0.472]			-0.002 (0.002) [0.215]
SC × Baseline: Alcoholic drinks past week§			0.005 (0.014) [0.721]			-0.018 (0.020) [0.394]
SC × Baseline: Hours worked own farm			0.001 (0.001) [0.447]			0.002 (0.001) [0.091]
SC × Baseline: Hours non-farm wage work			0.000 (0.001) [0.553]			0.002 (0.001) [0.083]
SC × Baseline: Hours engaged in housework			0.000 (0.001) [0.893]			-0.003 (0.002) [0.268]
Constant	0.939 (0.016) [0.001]	0.937 (0.016) [0.001]	0.937 (0.017) [0.000]	0.866 (0.031) [0.000]	0.863 (0.034) [0.001]	0.858 (0.029) [0.000]
Mean of dep. variable		0.935			0.870	
Observations	525	525	525	525	525	525

Notes: Treat=Assigned to treatment. SC=Spillover control. The reference group is those in the pure control. Household-level demographic variables created by adding husband and wife characteristics (e.g., “Household age” is combined husband and wife ages). §Variable measures the most each individual drank in one day over the past week prior to the baseline survey, and for the household-level, summed between both spouses. Heteroskedasticity-robust standard errors are clustered at the village level and wild bootstrap p-values are shown in brackets below estimated coefficients and standard errors.

TABLE A6
ALTERNATIVE ESTIMATIONS: SPOUSE-REPORTED ALCOHOL USE PAST WEEK

	Extensive margin: Short run						Extensive margin: Medium run					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Assigned to treatment	-0.173 (0.056) [0.012]	-0.177 (0.048) [0.002]	-0.287 (0.069) [0.000]	-0.305 (0.057) [0.000]	-0.060 (0.056) [0.340]	-0.045 (0.049) [0.413]	-0.123 (0.042) [0.018]	-0.122 (0.037) [0.007]	-0.177 (0.050) [0.005]	-0.187 (0.044) [0.002]	-0.070 (0.053) [0.270]	-0.052 (0.051) [0.358]
Spillover control	-0.101 (0.057) [0.135]	-0.092 (0.049) [0.108]	-0.136 (0.067) [0.083]	-0.116 (0.058) [0.082]	-0.065 (0.060) [0.310]	-0.068 (0.052) [0.239]	-0.066 (0.043) [0.164]	-0.054 (0.038) [0.194]	-0.128 (0.057) [0.031]	-0.109 (0.051) [0.041]	-0.004 (0.058) [0.944]	0.001 (0.056) [0.995]
Constant	0.474 (0.050) [0.000]	0.474 (0.042) [0.001]	0.779 (0.057) [0.000]	0.782 (0.046) [0.000]	0.169 (0.049) [0.022]	0.165 (0.044) [0.000]	0.500 (0.030) [0.000]	0.499 (0.027) [0.001]	0.856 (0.025) [0.000]	0.859 (0.015) [0.000]	0.144 (0.045) [0.007]	0.138 (0.045) [0.000]
Sample	M,W	M,W	M	M	W	W	M,W	M,W	M	M	W	W
Estimation [†]	ANA	ACV	ANA	ACV	ANA	ACV	ANA	ACV	ANA	ACV	ANA	ACV
Pure control mean	0.474	0.474	0.779	0.779	0.169	0.169	0.500	0.500	0.856	0.856	0.144	0.144
b[Treat]=b[SC] (p-val)	0.038	0.006	0.007	0.000	0.862	0.440	0.112	0.017	0.342	0.094	0.100	0.158
Treatment lower bound	-0.186	-0.187	-0.308	-0.323	-0.064	-0.049	-0.125	-0.125	-0.183	-0.195	-0.070	-0.052
Treatment upper bound	-0.159	-0.164	-0.282	-0.300	-0.037	-0.026	-0.122	-0.121	-0.176	-0.186	-0.070	-0.052
Treat: MHT (p-val) [‡]	0.086	0.051	0.018	0.008	0.509	0.597	0.055	0.038	0.015	0.000	0.532	0.624
Observations	982	982	491	491	491	491	874	874	437	437	437	437

	Heavy drinker: Short run						Heavy drinker: Medium run					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Assigned to treatment	-0.142 (0.036) [0.000]	-0.145 (0.030) [0.000]	-0.259 (0.062) [0.000]	-0.277 (0.054) [0.000]	-0.026 (0.021) [0.278]	-0.021 (0.019) [0.324]	-0.114 (0.033) [0.006]	-0.112 (0.034) [0.008]	-0.180 (0.056) [0.007]	-0.192 (0.058) [0.007]	-0.047 (0.038) [0.270]	-0.034 (0.035) [0.487]
Spillover control	-0.065 (0.036) [0.108]	-0.058 (0.030) [0.071]	-0.091 (0.068) [0.206]	-0.071 (0.059) [0.252]	-0.039 (0.020) [0.095]	-0.040 (0.018) [0.042]	-0.069 (0.038) [0.085]	-0.057 (0.037) [0.138]	-0.147 (0.060) [0.021]	-0.125 (0.058) [0.037]	0.009 (0.045) [0.877]	0.013 (0.042) [0.802]
Constant	0.312 (0.027) [0.002]	0.312 (0.020) [0.002]	0.565 (0.044) [0.002]	0.568 (0.037) [0.000]	0.058 (0.017) [0.001]	0.057 (0.015) [0.004]	0.367 (0.020) [0.000]	0.366 (0.022) [0.003]	0.662 (0.027) [0.000]	0.666 (0.031) [0.000]	0.072 (0.036) [0.032]	0.068 (0.035) [0.015]
Sample	M,W	M,W	M	M	W	W	M,W	M,W	M	M	W	W
Estimation [†]	ANA	ACV	ANA	ACV	ANA	ACV	ANA	ACV	ANA	ACV	ANA	ACV
Pure control mean	0.312	0.312	0.565	0.565	0.058	0.058	0.367	0.367	0.662	0.662	0.072	0.072
b[Treat]=b[SC] (p-val)	0.015	0.002	0.010	0.001	0.392	0.211	0.228	0.081	0.607	0.258	0.052	0.069
Treatment lower bound	-0.151	-0.152	-0.274	-0.290	-0.027	-0.022	-0.115	-0.114	-0.185	-0.196	-0.047	-0.034
Treatment upper bound	-0.124	-0.126	-0.247	-0.265	-0.001	0.004	-0.112	-0.111	-0.178	-0.190	-0.047	-0.034
Treat: MHT (p-val) [‡]	0.009	0.000	0.004	0.000	0.501	0.631	0.015	0.019	0.021	0.020	0.420	0.481
Observations	982	982	491	491	491	491	874	874	437	437	437	437

Notes: [†]ANOVA (ANA) and ANCOVA (ACV) estimations are calculated using linear probability models. Dependent variable for extensive margin (top panel) is whether any alcohol was consumed in the week prior to the survey (spouse-reported). “Baseline: Alcohol past week (binary)” measures whether the respondent drank in the past week in the baseline survey (self-reported). Heavy drinker (bottom panel) defined as consumption of at least 15 (8) US standard drink equivalents per week for men (women). Sample refers to pooled gender (M,W), Men (M), or Women (W). Demeaned variables included but not shown: control for male drinker in pooled gender estimates, baseline alcohol consumption. “b[Treat]=b[Sp. control]” reports the p-value for a Wald test on whether the estimated coefficient of the treatment is statistically different from the estimated coefficient of the spillover control. I calculate upper and lower bounds following Lee (2009) and McKenzie (2017). Medium-run estimations only include sample that was nonattreated in short-run (i.e., excludes those found and surveyed in 2022 but not 2021). [‡]I calculate multiple hypothesis test p-values following List, Shaikh and Xu (2019) and Steinmayr (2020), which control for the familywise error rate using bootstrapped resampling and allows for p-values to be correlated. Heteroskedasticity-robust standard errors are clustered at the village level, and wild bootstrap p-values shown in brackets below estimated coefficients and standard errors.

TABLE A7
AGRICULTURAL OUTCOMES - BASELINE ANALYSIS

	Harvest Value 1000s KSh			Inorganic Fertilizer Kg			Organic Inputs Kg			Hired Labor (Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Assigned to treatment	0.034 (0.225)	0.047 (0.152)	0.043 (0.138)	-0.020 (0.190)	-0.025 (0.114)	-0.051 (0.109)	-0.815 (0.304)	-0.775 (0.322)	-0.789 (0.285)	-0.027 (0.178)	-0.020 (0.184)	-0.011 (0.218)
Spillover control	0.011 (0.204)	0.024 (0.139)	0.044 (0.128)	-0.042 (0.192)	-0.037 (0.114)	-0.125 (0.108)	-1.185 (0.332)	-1.157 (0.340)	-1.483 (0.299)	0.072 (0.175)	0.082 (0.184)	0.041 (0.212)
Constant	3.505 (0.178)	3.376 (0.126)	3.364 (0.115)	4.463 (0.130)	4.301 (0.078)	4.333 (0.071)	5.103 (0.201)	5.052 (0.197)	5.073 (0.180)	2.019 (0.106)	1.883 (0.122)	1.896 (0.136)
Estimation†	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV
Pure control mean	33.277	33.277	33.277	86.757	86.757	86.757	164.545	164.545	164.545	7.532	7.532	7.532
Observations	491	491	491	491	491	491	491	491	491	491	491	491

Notes: [†]ANOVA (ANA), ANCOVA (ACV), and ANHECOVA (AHV) estimations after Ye et al. (2023) are calculated using Poisson QMLE (Correia, Guimarães and Zylkin, 2020). Coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Outcomes are annual (two season) totals. Demeaned variables included but not shown: plot size and its square in ANCOVA estimates, which are additionally interacted with treatment group assignment in ANHECOVA estimates. Heteroskedasticity-robust standard errors are clustered at the village level.

TABLE A8
ALTERNATIVE ESTIMATIONS: HOURS WORKED YESTERDAY & WAGES PAST
MONTH

	Hours worked own farm			Hours worked self employ			Hours worked for wages		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Assigned to treatment	0.024 (0.126)	-0.049 (0.179)	0.139 (0.180)	0.177 (0.163)	0.221 (0.175)	0.111 (0.231)	-0.244 (0.152)	-0.155 (0.150)	-0.467 (0.295)
Spillover control	0.019 (0.107)	-0.035 (0.163)	0.099 (0.176)	0.247 (0.187)	0.328 (0.220)	0.094 (0.216)	-0.349 (0.154)	-0.298 (0.149)	-0.490 (0.314)
Constant	0.666 (0.090)	0.893 (0.149)	0.372 (0.143)	0.461 (0.131)	0.666 (0.121)	0.203 (0.161)	0.784 (0.086)	1.114 (0.074)	0.287 (0.220)
Sample	M,W	M	W	M,W	M	W	M,W	M	W
Pure control mean	1.946	2.443	2.443	1.586	1.946	1.225	2.189	3.046	1.332
Treat: MHT (p-val)‡	0.995	0.995	0.959	0.875	0.884	0.994	0.670	0.959	0.630
Observations	885	442	443	885	442	443	885	442	443

	Any wage earnings			Any non-farm wage earnings			Total labor earnings†		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Assigned to treatment	-0.056 (0.048)	0.001 (0.040)	-0.109 (0.068)	-0.012 (0.035)	-0.002 (0.049)	-0.018 (0.046)	-0.120 (0.114)	-0.088 (0.116)	-0.238 (0.209)
Spillover control	-0.076 (0.048)	-0.031 (0.044)	-0.123 (0.067)	-0.006 (0.036)	-0.013 (0.044)	-0.002 (0.057)	-0.082 (0.154)	-0.114 (0.142)	0.029 (0.269)
Constant	0.696 (0.040)	0.829 (0.030)	0.564 (0.055)	0.411 (0.024)	0.636 (0.032)	0.186 (0.038)	1.994 (0.072)	2.485 (0.066)	0.987 (0.178)
Sample	M,W	M	W	M,W	M	W	M,W	M	W
Pure control mean	0.696	0.829	0.564	0.696	0.636	0.186	7.344	12.003	2.684
Treat: MHT (p-val)‡	0.809	0.999	0.556	0.995	0.999	0.995	0.955	0.987	0.875
Observations	885	442	443	885	442	443	885	442	443

Notes: ANOVA estimations. Top panel estimations and estimates for total labor earnings in bottom panel use Poisson QMLE (Correia, Guimarães and Zylkin, 2020). These coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Sample refers to pooled gender (M,W), Men (M), or Women (W). †I calculate “Total earnings from labor” as earnings from self employment (calculated after Agness et al. (2025)) and actual market wages over the past 30 days in thousands of Kenyan Shillings. “Hours worked” outcomes use data collection adapted from Field et al. (2023), in which respondents were asked to divide up their last 24 hours into nine categories, including sleeping, working on own farm, self employment, unpaid household work, etc. ‡I calculate multiple hypothesis test p-values here following Romano and Wolf (2016) and Clarke, Romano and Wolf (2020), which control for the familywise error rate and uses bootstrapped resampling to allow for dependence among the p-values. Heteroskedasticity-robust standard errors are clustered at the village level.

TABLE A9
RANDOMIZATION INFERENCE

Outcome Variable	Cluster robust p-value	Rand. Inference p-value
Positive value on breathalyzer	0.000	0.007
Spouse reported alcohol consumed past 7 days SR survey (male - binary)	0.000	0.000
Spouse reported heavy drinking SR survey (male - binary)	0.000	0.000
Spouse reported alcohol consumed past 7 days MR survey (male - binary)	0.001	0.002
Spouse reported heavy drinking MR survey (male - binary)	0.005	0.003
Harvest value 1000s KSh	0.005	0.013
Organic fertilizer Kg	1.840	0.885
Inorganic fertilizer Kg	0.006	0.008
Hired labor (quantity)	0.021	0.022

Notes: Table reports results of randomization inference p-value compared to the original cluster-robust p-value for the primary outcomes in this study. I re-conduct the two stages of the randomization (village level, then household level) 1000 times and the primary results re-estimated for each new iteration to generate placebo estimates. The randomization inference p-value is then calculated as the proportion of times the placebo estimate has a greater magnitude than the original ITT estimate. Specifications correspond to ANHECOVA estimates in Tables 2, 4, and 5. Cluster robust standard errors for alcohol outcomes are calculated using a wild bootstrap.

TABLE A10
MECHANISMS

1. Intervention → Lower Alcohol Consumption	
Potential Mechanism	Evidence
[1.1] The individual learns that returns to consumption are low/negative.	Qualitative data from counselor reports (Appendix C).
[1.2] The individual updates their own ability to stop drinking.	Qualitative data from counselor reports (Appendix C).
[1.3] Spouse or other members of community help individual commit to not drinking/hold them accountable for not drinking.	Analysis of social network data (Murphy, 2023).
[1.4] If drinking social, peer effects reduce need/desire to drink.	Analysis of social network data. (Murphy, 2023).
[1.5] The program affects religious attitudes, increasing self-control.	Empirical analysis discounts plausibility of this potential mechanism. (?).
[1.6] Withdrawal medication given by the nurse decreases urge to drink.	Statistics show more than 10% of those in treatment received withdrawal medication.
2. Lower Alcohol Consumption → Positive Economic Outcomes	
[2.1] Reallocation of spending from alcohol to productive inputs or hired labor.	Empirical analysis of expenditure data (Table 7).
[2.2] Improved health leads to greater labor productivity.	Cannot observe directly, but lower spending on healthcare provides suggestive evidence (Table 7).
[2.3] More clearheaded and able to make smarter choices about inputs.	Possible mechanism, but not able to conclude based on available data.
[2.4] More willingness to take risks and therefore more willing to use fertilizer.	Possible mechanism, but not able to conclude based on available data.
3. Intervention → Positive Economic Outcomes	
[3.1] Improved relationship with spouse leads to happier individual, more willing to work, better decision-making.	Results from Castilla, Aqeel and Murphy (2025) show decrease in intimate partner violence.
[3.2] Counseling leads to happier individuals in general.	Results from Castilla, Aqeel and Murphy (2025) show decrease in intimate partner violence.
[3.3] Conversations with peers from group workshops lead to sharing of productive agricultural information.	Analysis of social network data (Murphy, 2023).
[3.4] Medication/vitamins make individual healthier, or nurse visits allow other health problems to be addressed.	Cannot observe directly, but lower spending on healthcare provides suggestive evidence (Table 7).
[3.5] Incentive payments lead to more resources that can be invested in agriculture.	No relationship between incentive payments received and agricultural outcomes (Appendix Table A11).
4. Intervention → Positive Economic Outcomes → Lower Alcohol Consumption	
[4.1] Recommendations by counselors to use work on farm as way to control urge to drink.	Qualitative data from counselor reports (Appendix C). Empirical analysis is inconclusive (Tables A12 and A13).

TABLE A11
AGRICULTURAL OUTCOMES: INCENTIVE PAYMENTS

	Harvest Value 1000s KSh			Inorganic Fertilizer Kg			Organic Inputs Kg			Hired Labor (Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Incentive payments received (1000s KSh)	0.148 (0.194)	-0.041 (0.132)	-0.032 (0.130)	0.020 (0.213)	0.015 (0.078)	-0.000 (0.088)	-0.298 (0.238)	-0.262 (0.245)	-0.440 (0.253)	-0.231 (0.187)	-0.301 (0.230)	-0.290 (0.206)
Assigned to treatment	0.183 (0.207)	0.338 (0.133)	0.304 (0.125)	0.202 (0.264)	0.209 (0.106)	0.270 (0.127)	-0.097 (0.382)	-0.030 (0.421)	-0.047 (0.387)	0.396 (0.262)	0.438 (0.278)	0.655 (0.268)
Spillover control	0.129 (0.199)	0.169 (0.063)	0.086 (0.072)	-0.001 (0.186)	-0.039 (0.077)	0.024 (0.089)	-0.467 (0.312)	-0.332 (0.331)	-0.439 (0.272)	0.128 (0.193)	0.072 (0.160)	0.156 (0.185)
Constant	3.584 (0.162)	3.380 (0.048)	3.410 (0.059)	4.138 (0.134)	3.935 (0.064)	3.875 (0.068)	4.785 (0.218)	4.639 (0.216)	4.710 (0.170)	1.907 (0.129)	1.692 (0.112)	1.486 (0.138)
Estimation [†]	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV
Pure control mean	36.171	36.171	36.171	63.282	63.282	63.282	118.915	118.915	118.915	6.944	6.944	6.944
Observations	448	448	448	448	448	448	448	448	448	448	448	448

Notes: Incentive payments are in 1000s of KSh. 161 households received incentive payments from lotteries of between 500 and 2000 KSh, with the average amount, conditional on receiving any, of 1000 KSh. [†]ANOVA (ANA), ANCOVA (ACV), and ANHECOVA (AHV) estimations after Ye et al. (2023) are calculated using Poisson QMLE (Correia, Guimarães and Zylkin, 2020). Coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Outcomes are annual (two season) totals. Agricultural price data in 2022 is deflated to 2020 levels using Kenya Agricultural PPI (allias?). Demeaned variables included in ANCOVA but not shown: baseline of outcome, an indicator for whether farmers had fully completed the 2022 long rains harvest (see text for additional information on this variable), plot area and its square. These variables are interacted with treatment group assignment in ANHECOVA estimations. Heteroskedasticity-robust standard errors clustered at the village level.

TABLE A12
AGRICULTURAL OUTCOMES: COUNSELOR NOTES

	Harvest Value 1000s KSh			Inorganic Fertilizer Kg			Organic Inputs Kg			Hired Labor (Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Counselor notes show discussion of agriculture	0.022 (0.219)	-0.080 (0.175)	-0.073 (0.179)	0.082 (0.168)	0.127 (0.119)	0.101 (0.121)	-0.544 (0.409)	-0.622 (0.394)	-0.904 (0.432)	0.131 (0.226)	0.469 (0.172)	0.331 (0.179)
Assigned to treatment	0.299 (0.213)	0.341 (0.124)	0.312 (0.143)	0.178 (0.227)	0.161 (0.106)	0.219 (0.122)	-0.107 (0.370)	0.025 (0.396)	-0.052 (0.320)	0.144 (0.222)	-0.034 (0.226)	0.251 (0.221)
Spillover control	0.129 (0.199)	0.169 (0.063)	0.086 (0.072)	-0.001 (0.186)	-0.039 (0.077)	0.024 (0.089)	-0.467 (0.312)	-0.329 (0.333)	-0.439 (0.272)	0.128 (0.193)	0.065 (0.158)	0.156 (0.185)
Constant	3.584 (0.162)	3.381 (0.048)	3.410 (0.059)	4.138 (0.134)	3.933 (0.064)	3.875 (0.068)	4.785 (0.218)	4.633 (0.217)	4.710 (0.170)	1.907 (0.129)	1.683 (0.109)	1.486 (0.138)
Estimation [†]	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV
Pure control mean	36.171	36.171	36.171	63.282	63.282	63.282	118.915	118.915	118.915	6.944	6.944	6.944
Observations	448	448	448	448	448	448	448	448	448	448	448	448

Notes: Counselor disc. farming is equal to “1” if the counselor recorded ever discussing farm/agricultural work with the household in their reports. [†]ANOVA (ANA), ANCOVA (ACV), and ANHECOVA (AHV) estimations after Ye et al. (2023) are calculated using Poisson QMLE (Correia, Guimarães and Zylkin, 2020). Coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Outcomes are annual (two season) totals. Agricultural price data in 2022 is deflated to 2020 levels using Kenya Agricultural PPI (*(alias?)*). Demeaned variables included in ANCOVA but not shown: baseline of outcome, an indicator for whether farmers had fully completed the 2022 long rains harvest (see text for additional information on this variable), plot area and its square. These variables are interacted with treatment group assignment in ANHECOVA estimations. Heteroskedasticity-robust standard errors clustered at the village level.

TABLE A13
AGRICULTURAL BASELINE OUTCOMES: COUNSELOR NOTES

	Harvest Value 1000s KSh			Inorganic Fertilizer Kg			Organic Inputs Kg			Hired Labor (Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Counselor notes show discussion of agriculture	0.198 (0.133)	0.165 (0.113)	0.151 (0.118)	0.109 (0.111)	0.033 (0.127)	0.028 (0.126)	0.302 (0.432)	0.345 (0.425)	0.354 (0.427)	-0.391 (0.286)	-0.461 (0.253)	-0.444 (0.259)
Assigned to treatment	-0.064 (0.214)	-0.032 (0.158)	-0.029 (0.150)	-0.073 (0.191)	-0.040 (0.142)	-0.064 (0.138)	-0.968 (0.338)	-0.946 (0.349)	-0.968 (0.336)	0.138 (0.213)	0.168 (0.221)	0.170 (0.241)
Spillover control	0.011 (0.204)	0.025 (0.139)	0.044 (0.128)	-0.042 (0.192)	-0.037 (0.114)	-0.125 (0.108)	-1.185 (0.332)	-1.156 (0.340)	-1.483 (0.299)	0.072 (0.175)	0.078 (0.184)	0.041 (0.212)
Constant	3.505 (0.178)	3.375 (0.126)	3.364 (0.115)	4.463 (0.130)	4.301 (0.078)	4.333 (0.071)	5.103 (0.201)	5.050 (0.198)	5.073 (0.180)	2.019 (0.106)	1.886 (0.120)	1.896 (0.136)
Estimation [†]	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV	ANA	ACV	AHV
Pure control mean	33.277	33.277	33.277	86.757	86.757	86.757	164.545	164.545	164.545	7.532	7.532	7.532
Observations	491	491	491	491	491	491	491	491	491	491	491	491

Notes: Baseline variables are outcomes. Counselor disc. farming is equal to “1” if the counselor recorded ever discussing farm/agricultural work with the household in their reports. [†]ANOVA (ANA), ANCOVA (ACV), and ANHECOVA (AHV) estimations after Ye et al. (2023) are calculated using Poisson QMLE (Correia, Guimarães and Zylkin, 2020). Coefficients can be interpreted as a percent change using the incidence rate ratio ($e^{\beta} - 1$). Outcomes are annual (two season) totals. Demeaned variables included in ANCOVA but not shown: plot area and its square. These variables are interacted with treatment group assignment in ANHECOVA estimations. Heteroskedasticity-robust standard errors clustered at the village level.

B Intervention Details, Participation Incentives, and Safety Protocols

1 Intervention Stages

Stage 1: Contact and Screening: The team of counselors and social workers first worked with village guides to meet and introduce themselves to each treatment household. At this time, the team conducted an initial screening to determine the family structure and alcohol consumption pattern of the household. The household was then informed about and invited to the community workshops. At this time, the team also informed the participants about the financial incentives to attend (further discussed below). This stage lasted between two and three days per village.

Stage 2: Workshops and Discussions: In each village, the implementation team held three days of structured counseling and discussions at a central location (often a church or school, which was rented for the occasion). Only those assigned to the treatment were invited and allowed to attend these workshops. The social workers and counselors divided each day into three parts:

1. *Sharing of personal problems, slips, frustrations, as well as success and growth:* The first part involved brief presentations by each member in the group, with a focus on “personal problems, constraints, negative encounters, and pressures to use alcohol or drugs since the last group meetings.” Other members at this point shared how they coped with similar experiences, and the counselors underlined skills to overcome difficulties to achieve rehabilitation.
2. *Whole group discussion based on the topic of the day:* These topics varied, but included: coping with social/family pressure, identifying personal triggers, the value of productive activities to diminish urges to drink, understanding psychological dependence, etc. This segment included both presentation and discussion of the topic.
3. *Defining next steps and specific tasks to achieve rehabilitation:* At this stage, the counselors assigned specific tasks to individual members according to their specific challenges. Their efforts to complete these tasks and overcome the associated challenges serve as a basis for discussion at successive meetings.

These sessions were confidential – the group counselors and social workers never shared what was discussed at these meetings with those who were not in attendance. It was also voluntary – attendance was not required, and those in attendance were not required to speak. The workshops and discussions, however, provided a platform for mutual support through discussions with the group counselors and others in the community that had alcohol problems. These workshops were not necessarily held on consecutive days during the team’s time in a village, but were spread across five or more days to better accommodate participants’ schedules. If participants paid for transport to the workshop they were reimbursed for their travel, and all participants received lunch while there.

Stage 3: Individual and Spousal Counseling: Each intervention team visited the treatment households at least eight times to complete individual and spousal sessions consisting of psychosocial

counseling as described above (using a MET framework). As reported in notes recorded by counselors as part of this program, the counselors also would develop both individual and joint spousal strategies to help control alcohol urges, for example, to focus on agricultural work as a distraction from drinking.²³ During this time, a staff nurse would visit the households to monitor withdrawal symptoms, and if needed, would prescribe vitamins and/or medication.

2 *Incentives for Participation*

The study included incentives for treatment individuals to participate in the program. While direct costs of attendance, such as transportation to the workshops, were reimbursed, there was also an opportunity cost to the treatment individuals for participation. The project therefore provided the opportunity to receive incentives if an individual participated. If an individual attended at least two workshop sessions, they were entered into a lottery to win 500 KSh (approximately one day's wage in the area). Each individual's chance of winning the 500 KSh was 50 percent. A second lottery was held after the completion of the counseling stage. If an individual had attended at least four sessions, they were entered to win another 500 KSh. The probability of winning was again 50 percent, and the outcome of the second lottery was independent of the first lottery. Therefore, the expected value of winnings for an individual who participated in at least half of both stages was 500 KSh. The results in Appendix Table A2, which do not show a large drop-off in counseling sessions attended after the minimum number to be entered into the lottery, suggest that the primary driver of participation was not this incentive, but a demand for sobriety.

3 *Safety Protocols*

When a person is an alcoholic, the body becomes chemically dependent on ethanol. This means that cessation or significant reductions in alcoholic drink can trigger withdrawal symptoms. Usually, these symptoms are relatively mild. However, in rare cases, serious and adverse medical effects can occur from reductions in alcohol consumption. This includes *delirium tremens*, which can lead to hallucinations, seizures, and death. Because this study's primary goal was reduction in alcohol consumption in the treated sample, this study took precautions on two fronts. First, it limited the sample to those 55 years old or younger and excluded those with known comorbidities or underlying diseases from participating, as older age and other underlying health conditions are known to increase the chance of onset of delirium tremens (Wetterling et al., 1994; Palmstierna, 2001; Jesse et al., 2017). Second, it employed a full-time nurse practitioner who visited all treated households, provided vitamins to all, and prescribed withdrawal medication to those experiencing symptoms. The nurse and counselors recorded 44 individuals (out of 400 assigned to the treatment) as requiring medication for withdrawal symptoms. Those with minor symptoms that did not require medication (e.g., headaches, light nausea, etc.) are not included in this total.

²³Appendix C contains anonymized excerpts from counselor notes relating to this strategy.

C Counselor Notes

In this program, counselors took detailed reports during visits to households.

Examples of Mechanisms 1.1 and 1.2: Individual learns that returns to consumption are low/negative; Individual updates beliefs in their own ability to stop drinking: •“M— feels proud of having being able to stay for almost a week now without alcohol. He says he is out for change and is not ready to relapse after realizing how harmful alcohol can be. We pondered on what he thinks about his drinking. M— says that he was having a problem with alcohol probably because he had no information about drugs and alcohol. He thought alcohol was the only way out to relieve stress but realized he was all wrong.” •“I clarified to W— that the decision to quit drinking chang’aa basically relied upon him and not any other person, neither a counselor nor the wife should push him to quit drinking. I further personalized the risks of drinking chang’aa to W—. These included death from effects caused by drinking that includes, liver cirrhosis, high blood pressure, heart attack and stroke. Poverty as most of the money he has been spending on buying chang’aa and not development.” •“C— has reduced alcohol intake from daily consumption to a cup after every two days. I congratulated him for the step made. He has formed a group of friends to sensitize on effects of alcohol. He has demonstrated strong will to stop taking alcohol.” •“Personalizing the risks alcohol has had on his life. After a recap of these topics covered during the training, E— was able to identify the following as his social economic effects: Marital conflict, R— says they used to fight a lot but it never got physical. During his drinking days he was unemployed, therefore getting employment was least of his concerns his major concern being where he would get his next drink. R— says the fights would happen in the presence of their children therefore E— wasn’t setting a good example for them. R— also says he could not save up to start a business because he would spend everything on alcohol. E— says that he is now fully involved in his business, he understands that alcohol cannot go hand in hand with business and this is the major reason he reduced his consumption.” •“After the training, this was an eye opener for him. He decided to quit alcoholism. His drinking habits have led to economic instability. All the money he gets from his Boda-Boda business gets wasted on alcohol. He admitted he is an addict, which has also caused domestic violence in his marriage” •“During the three day training organized prior to the family therapy, M— came to know of the adverse health effects of alcohol. This changed M—’s perspective of alcohol, informing her decision to recover from addiction. M— has observable and elaborate plan to quit alcoholism.” •“To him,it has been his toxic friends,life stresses and even poverty that has made him abuse alcohol.He says he has sold almost everything in the house and that he does not own even a bed to lie on.He regrets how much alcohol has costed him and the number of times he slept in a police cell.M— breaks down while at it and shows how remorseful he is to his dear wife.” •“The client has also realised that he uses a lot of money in buying alcohol and can use this money to pay school fees for his children and also put savings that can help start a business for his wife.” •“Alcoholism has reduced their economic growth.He uses almost all his finances to support changaa and Busaa drinking.He admitted being an addict. His family has suffered from domestic violence,incurred injuries and loss of valuable properties to alcohol.He is already in contemplation stage. He chose to quit drinking alcohol and with my support I expressed empathy,praise,show validation and encouragement during the contemplation stage and guided clients towards examining the benefits of the barriers to change.”

Examples of Mechanism 4.1: Recommendations by counselors to use work on farm as a way to control urge to drink: •“Taking up farm responsibilities after work to keep them busy so as to avoid drinking buddies;” •“He identified idleness as an obstacle to his recovery journey. I encouraged him to involve himself in farming and other things that can benefit his family;” •“As a family they have agreed to concentrate in their half acre farm to maximize the production. In my therapy session today I stressed on the need to have a short achievable economic plan to kill idleness.” •“His point of action included....Concentrate on his farm work that he abandoned because of alcoholism.” •“Involving herself in farming to forget the changaa intake;”

Other reports indicate success from this strategy. For example: • “B—— has been more focused on farm work.” • “Has stayed focused on Recovery by keeping busy with work and farm activities.” •“S—— has not used any alcoholic drink since the session started. He is quite committed to following the road to sobriety. He was at the farm when I arrived indicating that he is keeping busy and keeping the mind off alcohol;” •“R—— has been able to avoid toxic friends, concentrate on farm work, avoid free alcohol offers and unnecessary hangouts;” •“Has had a total behavior change as he helps a lot around the house and in the family farm which was not the case when he used to drink;” •“The client has made remarkable changes in that today he reported that he did not take any alcohol and was confirmed by his wife J——. He has been in line following to the latter [sic] the solutions he had come up with to abstain from taking a drink. He, for example has been keeping busy with his farm work engaging him so much hence abstinence;” •“P—— and E—— show satisfactory changes. The couple confirmed they have not taken any alcohol and they have engaged fully in their farming activities, avoided influential [sic] friends.”

Appendix References

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